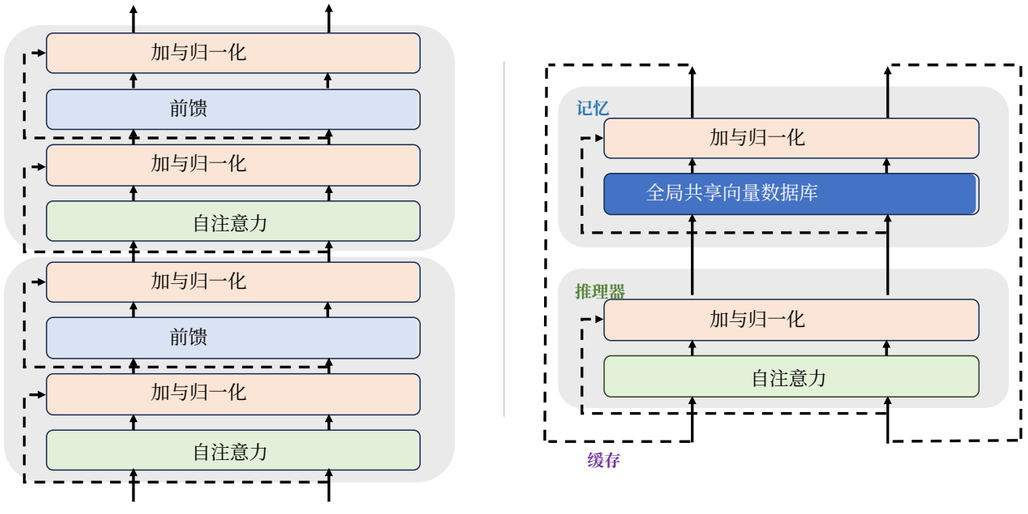
Intern-S2-Mobius：具有解耦知识与推理的基础模型

Intern-S2-Mobius团队，上海人工智能实验室

我们介绍了Mobius-v0，这是一种通过在基础模型内解耦知识向量与推理算子，内生支持反向残差连接和动态潜在推理的架构。Mobius由一个全局共享的记忆（存储知识向量）和多个推理器（迭代实现组合推理）组成。推理器以隐藏状态作为缓存和载体，反复查询记忆以获取所需的知识向量，同时知识被传回推理算子。通过这种知识-推理分离架构，Mobius实现了更好的知识压缩和推理效率。基于Mobius-v0架构：1）我们从头训练的7B模型在仅使用基线62.6%训练数据的情况下，达到了与7B Transformer基线相当的下游得分。2）我们的Intern-S2-Mobius从Qwen3.5-35B持续预训练而来，在实现相当下游得分的同时，提供了近 4× 倍的端到端推理加速。

1. 基础模型的发展瓶颈



(a) Transformer　　(b) Mobius

图1：Transformer与Mobius的对比。

Transformer [69] 至今仍是最普遍且最强大的架构范式，几乎渗透到每一个领域——从语言、视觉，到视频和科学计算 [4, 15, 16, 81]。在其演化轨迹中，出现了两个关键的研究方向。第一个方向在保持模型架构不变的同时，通过增加其他方面的投入来提升能力：扩展模型参数、扩展训练语料以及延长推理链，通常会使模型拥有更丰富的知识和

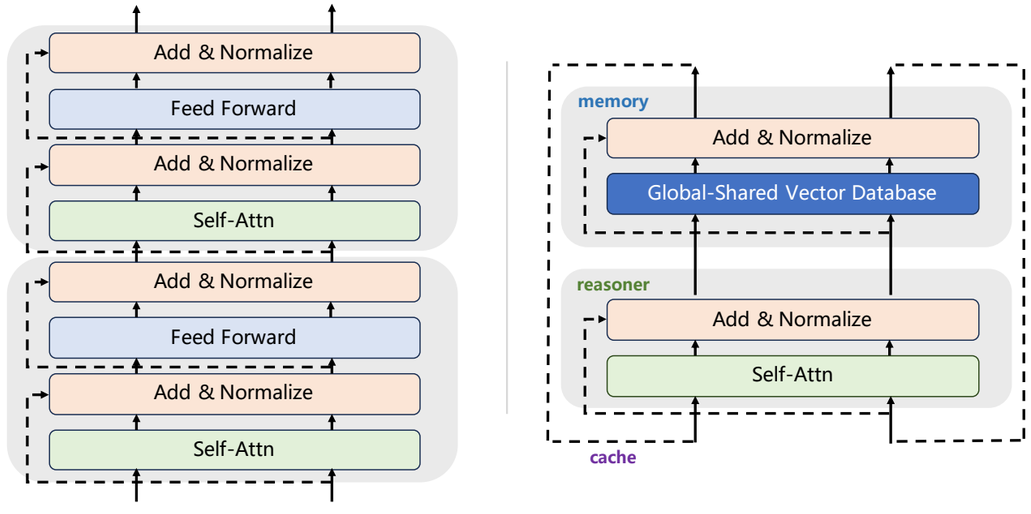
∗ 模型可在 https://huggingface.co/internlm/Intern-S2-Mobius获取

Intern-S2-Mobius: Foundation Model with Decoupled Knowledge and Reasoning

Intern-S2-Mobius Team, Shanghai AI Laboratory

We introduce ***Mobius-v0***, an architecture that endogenously supports **Backward Residual Connection** and **Dynamic Latent Reasoning** by disentangling knowledge vectors and reasoning operators within the base model. Mobius comprises a globally shared Memory that stores knowledge vectors and multiple Reasoners that iteratively achieve compositional reasoning. Using hidden states as cache and carrier, reasoners repeatedly query memory for required knowledge-vectors, while the knowledge is transmitted back to reasoning operators. Through this knowledge-reasoning-separation architecture, Mobius achieves better knowledge compression and reasoning efficiency. Built upon Mobius-v0 architecture: 1) Our trained-from-scratch 7B model achieves similar downstream score as a 7B Transformer baseline with 62.6% of baseline's training data. 2) Our **Intern-S2-Mobius**, continually-pretrained from Qwen3.5-35B, achieves similar downstream score while delivering nearly 4× end-to-end inference speedup.

1. The development bottleneck of the Foundation Models



(a) Transformer　　(b) Mobius

Figure 1: The Comparison between Transformer and Mobius.

The Transformer [69] stands as the most prevalent and powerful architectural paradigm to date, permeating virtually every domain—from language and vision, to video and scientific computing [4, 15, 16, 81]. Along its evolutionary trajectory, two pivotal research directions have emerged. The first preserves the model architecture intact while increasing expenditure elsewhere to enhance capability: scaling model parameters, expanding training corpora, and elongating reasoning chains typically endow models with richer knowledge and the

∗ Model is available at https://huggingface.co/internlm/Intern-S2-Mobius

解决更复杂问题的能力 [33, 37, 45, 71]。第二个方向通过降低架构复杂度来减少训练和推理开销，以提高实用性：基于一种广泛持有的观点，即自注意力的二次复杂度构成了超长上下文的瓶颈，诸如SSM和GDN等线性注意力机制被提出 [10, 25, 38, 78]，其变体此后也被整合到主流当代架构中。

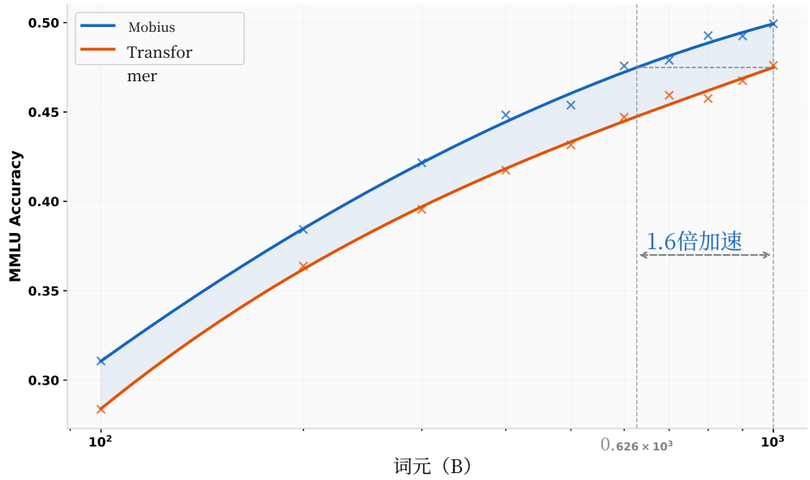


图2：Mobius与Transformer从头预训练的MMLU分数。

然而，这两种方法如今都已触及各自的极限。一方面，扩展数据和参数规模以高投资回报带来了日益强大的模型，但缩放定律已逐渐趋于平稳，正接近边际收益递减的临界点 [33, 37]。长思维链在数学、代码和物理等领域表现优异，然而无论问题难度如何，模型都倾向于产生冗长、偏离主题的输出——这一特征在人类社会通常被视为智力不足的表现 [49, 54]。另一方面，降低全注意力机制的计算复杂度确实带来了一定的效率提升，但日益明显的是，以牺牲相当一部分模型能力为代价所获得的效率改进，并未使模型真正足够经济以适用于商业部署 [3]。

面对这一人工智能发展的瓶颈，我们探索了一条替代路径：通过增加架构复杂度来提升模型智能的上限，从而降低端到端开销。为此，我们提出了Mobius，它将知识向量与推理算子之间的绑定关系解耦，从而构建了一个可供所有推理算子组访问的共享知识向量数据库。尽管这一共享向量数据库的庞大规模使得每次推理激活高度稀疏，带来了更大的内存访问压力和更低的单次前向效率，但我们发现，该实现相较于同等参数的Transformer架构，在保持等效推理精度的同时，实现了近 4×倍的端到端推理加速。

Mobius推理的效率提升主要源于两个来源：推理过程中更灵活的激活路径，以及更动态的潜在空间迭代，这两者共同赋予Mobius更高效、更简洁的输出模式。第一个优势源于，与Transformer中分层存储与前向残差连接的组合不同，Mobius的共享存储天然引入了反向残差连接。这一机制意味着，不仅浅层隐藏状态可以访问深层知识，深层隐藏状态也可以访问浅层知识，从而增强了跨层的组合泛化能力，进而加速了关键信息的综合。第二个优势源于Mobius天然引入了动态潜在推理：Transformer以词元作为信息传递的媒介，并且需要遍历所有层才能完成单个词元的推理，而Mobius可以在仅几层之内，针对完整知识库对潜在变量进行迭代和精炼，而无需像传统潜在推理那样进行多次全层迭代。这些潜在变量不与任何特定词元紧密绑定，只有在更深层才会同步解码多个词元。这种方法

capacity to tackle more intricate problems [33, 37, 45, 71]. The second reduces training and inference overhead by lowering architectural complexity to improve practicality: motivated by the widely held view that the quadratic complexity of self-attention constitutes a bottleneck for ultra-long contexts, linear attention mechanisms such as SSM and GDN have been proposed [10, 25, 38, 78], and their variants have since been integrated into mainstream contemporary architectures.

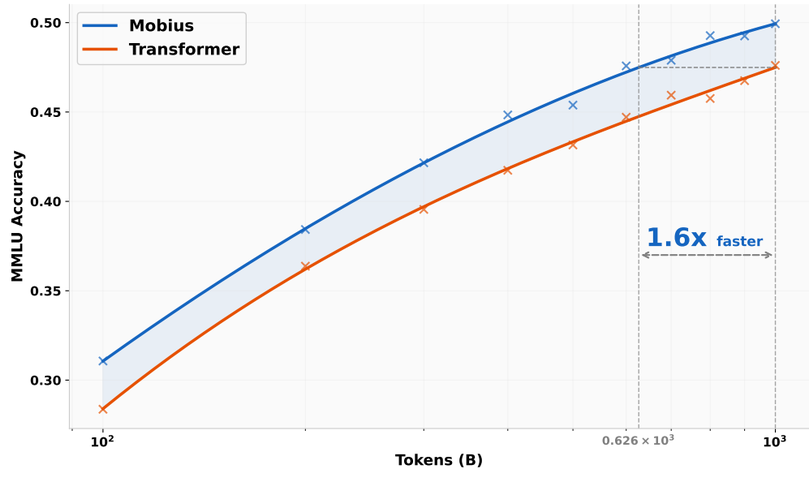


Figure 2: The MMLU score of Mobius and Transformer pre-trained from scratch.

Yet both approaches are now hitting their limits. On one hand, scaling data and parameters has yielded increasingly powerful models at high returns on investment, yet the Scaling Law has gradually plateaued and is approaching diminishing marginal returns [33, 37]. Long chains of thought have served us well in domains such as mathematics, code, and physics, yet models tend to produce verbose, tangential output regardless of problem difficulty—a trait typically regarded in human society as a sign of insufficient intelligence [49, 54]. On the other hand, reducing the computational complexity of full attention does bring certain efficiency gains, yet it has become increasingly apparent that the efficiency improvements achieved at the cost of sacrificing a substantial portion of model capability have not made models truly affordable enough for commercial deployment [3].

Confronted with this bottleneck in AI development, we pursue an alternative path: **increasing architectural complexity to raise the ceiling of model intelligence, thereby reducing the end-to-end overhead.** To this purpose, we propose the ***Mobius***, which decouples the binding between knowledge vectors and reasoning operators, thereby constructing a shared knowledge vector database accessible to all reasoning operator groups. Although the large scale of this shared vector database renders each inference activation highly sparse, introducing greater memory access pressure and lower per-pass forward efficiency, we find that this implementation achieves nearly 4× end-to-end inference speedup over the Transformer architecture with same parameters while maintaining equivalent reasoning accuracy.

The efficiency gains of Mobius reasoning stem primarily from two sources: a more flexible activation path during inference, and a more dynamic latent-space iteration, which together endow Mobius with a more efficient and concise output pattern. The first advantage arises because, unlike the combination of hierarchical storage and forward residual connections in Transformer, Mobius's shared storage natively introduces **Backward Residual Connection**. This mechanism means that not only can shallow hidden states access deep-layer knowledge, but deep hidden states can also access shallow-layer knowledge, which enhances the compositional generalization across different layers and thereby accelerates the synthesis of critical information. The second advantage arises because Mobius natively introduces **Dynamic Latent Reasoning**: whereas Transformer uses tokens as the medium of information transfer and requires traversing all layers to complete the inference of a single token, Mobius can iterate and refine latents against the full knowledge repository within just a few layers, rather than requiring multiple full-layer iterations as in traditional latent reasoning. These latents are not tightly bound to any specific token, and only at deeper layers are multiple tokens decoded synchronously. This approach

既提高了信息传递的密度，又动态地为不同词元分配不同的计算成本。最终，Mobius以显著更少的高质量词元完成相同的推理任务，实现了比Transformer高得多的端到端推理效率。

表1：通用与科学基准测试的性能对比。每行中的较高分数以粗体突出显示。

通用任务

基准	Intern-S2-Mobius-35B	Qwen3.5-35B
MMLU Pro	**89.05**	85.31
GPQA Diamond	**80.81**	80.24
IMO Bench	**81.25**	77.50
AIME 2026	**95.31**	92.08
HMMT 2026	**85.51**	78.50
UGD困难集	73.02	**78.02**
AMO	**58.00**	50.00
SimpleQA	**28.90**	21.39
HLE	19.11	**22.40**
平均分	**67.88**	65.05

科学任务

基准	Intern-S2-Mobius-35B	Qwen3.5-35B
生物学指令	**51.40**	3.77
分子指令	**45.73**	21.70
分子智能	**59.29**	29.13
平均分	**52.14**	18.20

2. 是什么启发了Mobius的设计?

在基于Transformer的架构中，前馈网络通常被认为负责知识存储 [20, 52], , 隐藏状态负责信息传输，而自注意力负责组合推理 [17, 69]。

同时，由于Transformer的层级结构，每一层的自注意力主要处理来自前一层的知识输入，并促使当前层的前馈网络产生下一层所需的输出。尽管残差连接建立了跨层更灵活的连接 [31, 34, 67], , 但只有较浅的隐藏状态能够访问深层知识，而深层的自注意力只能处理源自浅层的知识；模型仍然无法实现相反方向的信息传递。

为了构建反向的信息传递，当前模型主要依赖思维链 [26, 71]。每次前向传播后，模型生成一个新词元，用于在下一个推理步骤中激活知识。通过迭代词元生成，模型不断从前馈网络中提取有价值的知识，直到这些知识足以产生关键词元甚至最终答案。

仅使用层级结构、前向残差连接和词元介导的思维链来联合构建人工智能模型是低效的。为了使模型具备回答复杂问题的足够知识，大量计算被耗费在生成冗长且冗余的推理链上。为缓解这一问题，我们重新设计了模型架构并提出了Mobius。

2.1. Mobius 的第一项天赋——反向残差连接

残差连接已成为现代深度学习模型中最关键的组件之一 [31]。尽管看似微小，它们承担着在前向传播中传递层间信息、在反向传播中稳定梯度幅度的关键职责。 [32, 75] 尽管此后涌现出众多残差连接的变体,

both increases the density of information transfer and dynamically allocates varying computational costs to different tokens. Ultimately, Mobius completes the same reasoning task with markedly fewer high-quality tokens, achieving substantially higher end-to-end inference efficiency than Transformer.

Table 1: Performance comparison across general and scientific benchmarks. The higher score in each row is highlighted in **bold**.

General Tasks

Benchmark	Intern-S2-Mobius-35B	Qwen3.5-35B
MMLU Pro	**89.05**	85.31
GPQA Diamond	**80.81**	80.24
IMO Bench	**81.25**	77.50
AIME 2026	**95.31**	92.08
HMMT 2026	**85.51**	78.50
UGD hard	73.02	**78.02**
AMO	**58.00**	50.00
SimpleQA	**28.90**	21.39
HLE	19.11	**22.40**
AVG Score	**67.88**	65.05

Scientific Tasks

Benchmark	Intern-S2-Mobius-35B	Qwen3.5-35B
Biology-Instructions	**51.40**	3.77
Mol-Instructions	**45.73**	21.70
MolecularIQ	**59.29**	29.13
AVG Score	**52.14**	18.20

2. What inspired the design of Mobius?

In Transformer-based architectures, the Feed-Forward Network is conventionally regarded as responsible for knowledge storage [20, 52], Hidden States for information transmission, and Self-Attention for compositional reasoning [17, 69].

Concurrently, owing to the hierarchical structure of Transformer, Self-Attention at each layer primarily processes knowledge inputs received from the preceding layer and induces the FFN at the current layer to produce outputs required by the next layer. Although residual connections establish more flexible connections across layers [31, 34, 67], only shallow Hidden States can access deep-layer knowledge, while deep-layer Self-Attention can only process knowledge originating from shallow layers; the model remains incapable of information transfer in the opposite direction.

To construct information transfer in the reverse direction, current models predominantly rely on Chains of Thought [26, 71]. After each forward pass, the model generates a new token that serves to activate knowledge in the next inference step. Through iterative token generation, the model continuously extracts valuable knowledge from the FFN until this knowledge suffices to produce critical tokens or even the final answer.

The joint construction of artificial intelligence models using exclusively hierarchical structures, forward residual connections, and token-mediated chains of thought is inefficient. To equip models with sufficient knowledge for answering complex questions, an inordinate amount of computation is expended on producing lengthy, redundant reasoning chains. To mitigate this issue, we have redesigned the model architecture and propose Mobius.

2.1. Mobius' first innate talent —— Backward Residual Connection

Residual connections have become one of the most critical components in modern deep learning models [31]. Seemingly minor, they bear the crucial responsibility of transmitting inter-layer information during forward propagation and stabilizing gradient magnitudes during backpropagation. [32, 75] Although numerous variants

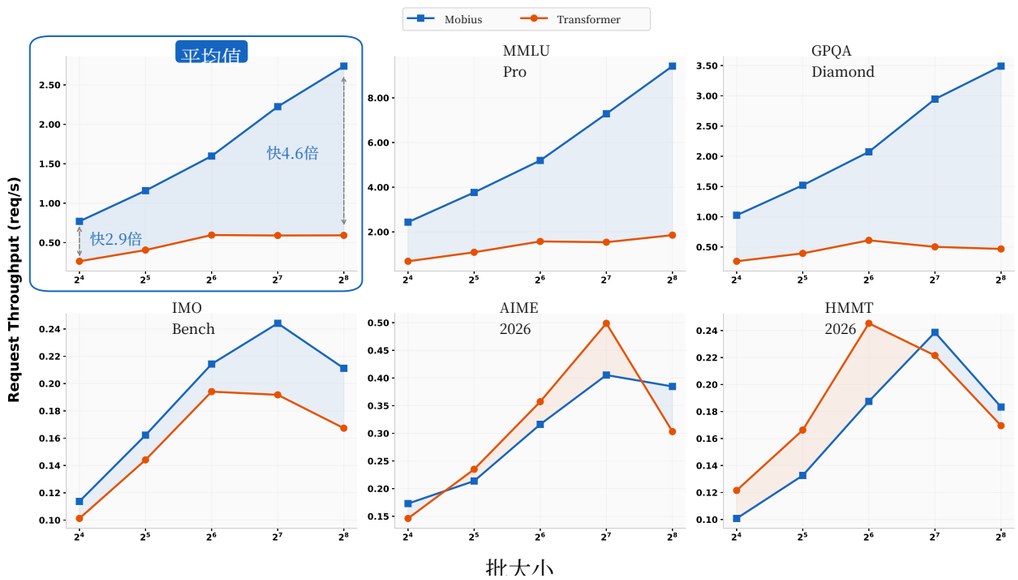


图3：由Qwen3.5持续预训练得到的Mobius的推理效率。

如 Hyper-Connection 和 Attention-Residual [39, 74, 80],，但迄今为止所有主流残差结构均保持其单向性。例如，在前向传播过程中，残差仅将浅层信息传递至更深层。 [34] 这种单向性带来了显著缺陷：若某些关键知识未能在浅层计算中被激活，模型在该轮推理中可能难以解码有价值的词元，只能通过思维链生成低信息词元以推进至下一步推理。

为缓解这一现象，有必要将残差连接改为双向——即在正向传播过程中使深层能够访问浅层知识。然而，直接实现此类反向残差连接对基础设施并不友好，因为它可能引入更复杂的计算图和更难并行化的异步问题。为此，我们选择以间接方式实现反向残差连接：不同层共享一个超大的统一知识库，使每一层都有机会访问模型内的全部知识。实验结果最终证实，这种形式的反向残差连接对提升模型的端到端推理效率具有实质性贡献。

2.2 Mobius的第二项天赋——动态潜在推理

长思维链已成为大语言模型推理中不可或缺的组成部分 [26, 45, 71]。借助这一技术，当代大语言模型已能解决涵盖数学、物理和代码生成等领域的广泛复杂推理问题。然而，长思维链的代价极为高昂。一方面，更长的推理链意味着需要生成更多词元，且生成成本随链长呈非线性增长。另一方面，当前模型在处理复杂问题时往往采用试错—纠错的方式，导致在扩展推理过程中产生大量令牌冗余。此外，这类模型即使面对简单问题也倾向于生成过于冗长的回答。这种冗长性部分源于上文讨论的残差连接，部分则源于当前推理的最小单元是词元——一种离散且信息密度较低的存储方式 [9]。

Mobius将深思熟虑、试错与改进等过程内化到连续向量的优化中，并使模型能够动态地为不同词元分配计算预算。这

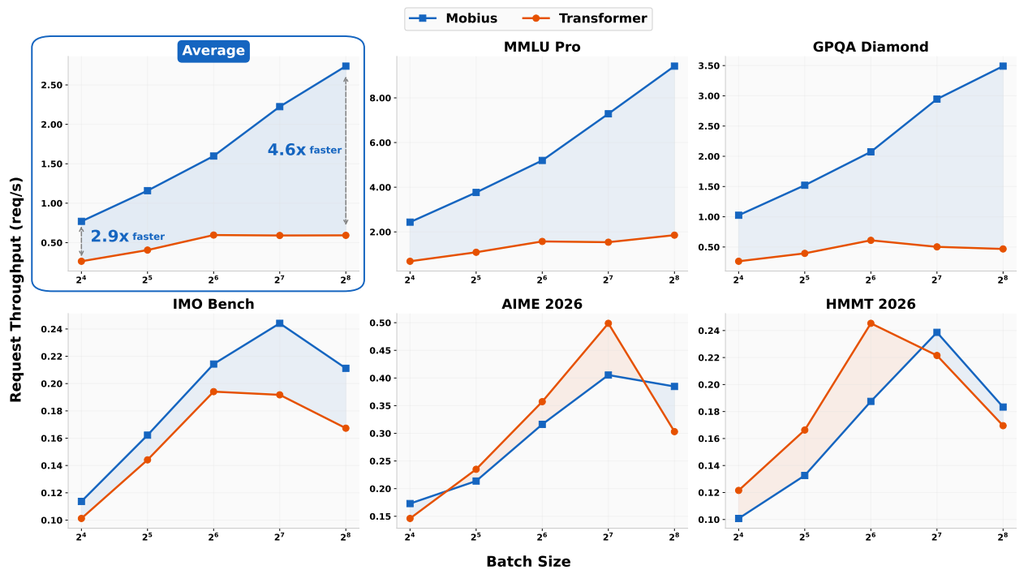


Figure 3: The inference efficiency of Mobius continually pre-trained from Qwen3.5.

of residual connections have since emerged, such as Hyper-Connection and Attention-Residual [39, 74, 80], all mainstream residual to date retain their unidirectional nature. For instance, during forward propagation, residual only convey shallow-layer information to deeper layers. [34] This unidirectionality entails significant drawbacks: if certain critical knowledge fails to be activated during shallow-layer computation, the model may struggle to decode valuable tokens in that reasoning round, and can only resort to generating low-information tokens via chains of thought to proceed to the next reasoning step.

To mitigate this phenomenon, rendering residual connections bidirectional is necessary—that is, enabling deep layers to access shallow-layer knowledge during forward propagation. However, a direct implementation of such backward residual connections is infrastructure-unfriendly, as it may introduce more complex computation graphs and harder-to-parallelize asynchrony. To this end, we opt for an indirect realization of backward residual connections: different layers share a single, oversized knowledge repository, granting every layer the opportunity to access all knowledge within the model. Empirical results ultimately corroborate that this form of backward residual connection contributes meaningfully to improving the model's end-to-end inference efficiency.

2.2. Mobius' second innate talent —— Dynamic Latent Reasoning

Long chains of thought (Long CoT) have become an indispensable component in large language model reasoning [26, 45, 71]. Empowered by this technique, contemporary large language models can now solve a wide spectrum of complex reasoning problems, spanning mathematics, physics, and code generation. Yet the cost of Long CoT is exorbitant. On one hand, longer reasoning chains entail the generation of more tokens, with generation cost growing non-linearly with chain length. On the other hand, current models frequently adopt a trial-and-error-and-correct approach when tackling complex problems, resulting in substantial token redundancy during extended reasoning. Moreover, such models tend to produce excessively verbose responses even for simple problems. This verbosity stems partly from the residual connections discussed above, and partly from the fact that the minimal unit of our current reasoning is the token—a discrete, low-information-density storage [9].

Mobius internalizes processes such as deliberation, trial-and-error, and refinement into the optimization of a continuous vector, and enables the model to dynamically allocate computational budget to different tokens. This

图4：从Qwen3.5持续预训练得到的Mobius的平均输出长度。

为模型创造了产生更少冗余词元、同时实现更高效、更智能推理的机会。Mobius可以被视为循环Transformer与扩散语言模型 [21, 43, 55], 的升级综合，采用了更高效的潜在推理和并行预测。首先，Mobius以更高的循环频率运行，用极少的层数完成一次表示迭代。其次，在推理过程中，Mobius通过更紧凑、高信息密度的连续向量同时获取多个词元的联合表示。这一方面缓解了并行化多个隐藏状态的压力，降低了循环过程中KV缓存管理的复杂度，并增强了向量迭代过程的连续可微性。同时，这种原生潜在推理赋予模型更动态、更不受约束的迭代过程，而不限制模型在特定迭代步骤解码固定数量的词元。

2.3. 一石二鸟——解耦知识向量与推理算子

总体而言，Mobius通过将知识存储模块（即前馈网络）与其逐层绑定关系解耦，并构建全局共享的知识向量数据库，从而实现了这两项内在能力。首先，这使得所有推理算子（即自注意力）能够读取全部知识体系。此外，自注意力得以在显著更少的层数内访问所有知识，并通过隐藏状态进行循环潜在推理，实现自适应多轮迭代，最终在一次爆发中输出精炼且高信息密度的词元。

在构建共享知识库时，我们选择了简单的水平拼接方式，因为前馈网络通常被视为键值知识向量池，而这种拼接方式保留了这些知识向量之间的对应关系。尽管这种简单的拼接带来了性能上的提升，但过多的参数激活会降低模型的训练和推理效率。因此，在更大的参数规模下，我们采用了类似于MoE [63] 的逐块划分技术来分割前馈网络，并在前向传播过程中进行稀疏激活。

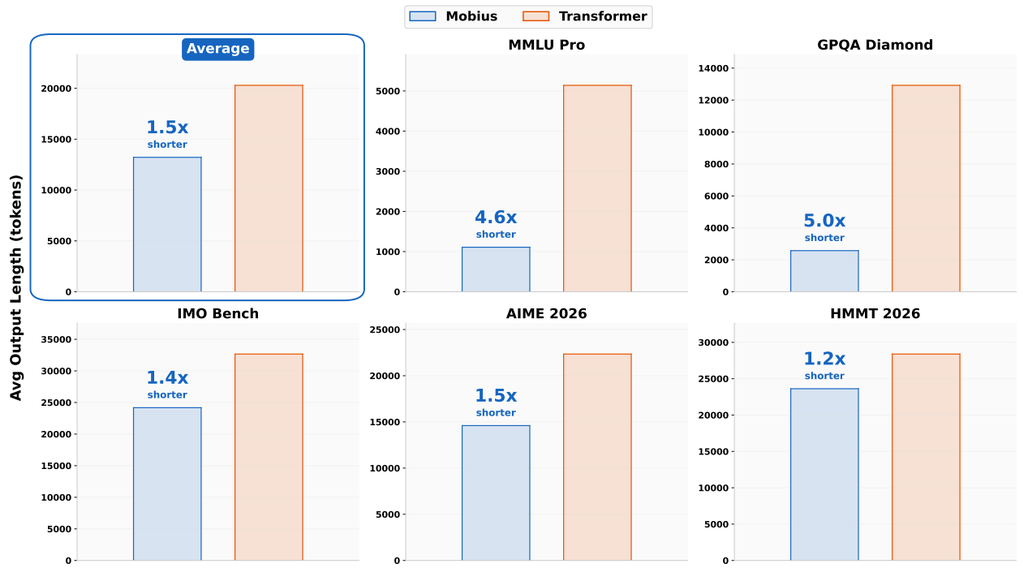


Figure 4: The average output length of Mobius continually pre-trained from Qwen3.5.

creates the opportunity for the model to produce fewer superfluous tokens while achieving more efficient and more intelligent reasoning. Mobius can be regarded as an upgraded synthesis of the Looped Transformer and the Diffusion Language Model [21, 43, 55], employing more efficient latent reasoning and parallel prediction. First, Mobius operates at a higher loop frequency, completing one representation iteration with extremely few layers. Second, during inference, Mobius acquires joint representations of multiple tokens simultaneously through more compact, high-information-density continuous vectors. This, on one hand, alleviates the pressure from parallelizing multiple hidden states, reduces the complexity of KV-cache management during the looping process, and enhances the continuous differentiability of the vector iteration process. Meanwhile, this native latent reasoning endows the model with a more dynamic and unfettered iteration process, without constraining the model to decode a fixed number of tokens at specific iteration steps.

2.3. One Stone Two Birds —— Disentangling Knowledge Vectors and Reasoning Operators

Overall, Mobius achieves these two innate talents by decoupling the knowledge storage module, the FFN, from its layer-wise binding, and constructing a globally shared knowledge-vector database. First, this grants all reasoning operators, the Self-Attention, read access to the entire body of knowledge. Further, the Self-Attention gains the opportunity to access all knowledge within significantly fewer layers, and performs adaptive multi-round iteration through recurrent latent reasoning via Hidden States, ultimately emitting refined, high-information-density tokens in a single burst.

In constructing the shared knowledge repository, we opted for a straightforward horizontal concatenation, as the FFN is conventionally regarded as a Key-Value Knowledge-Vector Pool, and this concatenation approach preserves the correspondence relationships among these knowledge vectors. Although such simple concatenation yields performance benefits, excessive parameter activation degrades the model's training and inference efficiency. Therefore, at larger parameter scales, we employ a block-wise partitioning technique analogous to MoE [63] to partition the FFN, with sparse activation during forward propagation.

表2：Intern-S2-Mobius-35B与Qwen3.5-35B在线性代数选择题上的步骤对齐比较。两个模型均选择了正确答案（选项C）。令牌计数使用Qwen3.5-35B分词器计算。Mobius以更少的词元完成了相同的推理步骤，这主要得益于该模型消除了重复推导和检查。

推理轨迹对比

对齐步骤	Intern-S2-Mobius-35B	Qwen3.5-35B
任务框架	**17个词元。** “用户希望我评估两个关于线性代数的陈述的真值。”	**20个词元。** “用户希望我评估两个陈述并判断其真值从选项列表中取值。”
陈述1	**178个词元。** “反例：设 $v_1 = (1, 0)$和 $v_2 = (2, 0)$。则 $2v_1 - v_2 = 0$，因此它们是线性相关的。……陈述1是错误的。”	**283个词元。** “考虑零向量。设 $v_1 = (1, 0)$ 和 $v_2 = (0, 0)$。……考虑两个平行向量。设 $v_1 = (1, 0)$ 和 $v_2 = (2, 0)$。……陈述1为假。”
陈述2	**142个词元。** “基是线性无关的生成集。由于 $\{v_1, \ldots, v_k\}$ 张成 V 且线性无关，它构成 V的一个基。……维数$(V) = k$。”	**197个词元。** “该集合构成 V的一个基，若它张成 V且线性无关。……因此，$\{v_1, \ldots, v_k\}$ 是 V. … 维数$(V) = k$。”
选项匹配	**22个词元。** “陈述1为假，陈述2为真。这与选项C匹配。”	**122个词元。** “组合为False，True。这对应选项C。”
重复推导与检查	—	**1,147个词元。** **“逐步推导：**……反例：设 $v_1 = (1, 0)$和 $v_2 = (2, 0)$。…… **最终检查：**……**如果** $k = 0$? ……可行。……选项C。”
可见的最终答案	**157个词元。** “向量 (1, 0) 和 (2, 0) 是线性相关，因为一个是另一个的标量倍数。. .. 这就是基的定义。. . . 答案：C ”	**595个词元。** **“步骤1：评估陈述1 . . .** $2v_1 - 1v_2 = (0, 0)$. .. . 陈述1为**错误。** **步骤2：评估陈述2 . . .** dim(V)等于 k。. .. 答案：C”
总计	**516个词元**	**2,364个词元**

3. Mobius的性能表现如何?

为验证Mobius的性能，我们进行了从头训练（TFS）和继续预训练（CPT）两组实验。在从头训练实验中，我们分别使用Mobius和Transformer在1TB词元上训练了7B-A1B混合专家模型，并比较了它们的MMLU分数。在继续预训练实验中，我们以Qwen3.5-35B-A3B为初始检查点，在1TB词元上继续预训练，随后进行了监督微调（SFT）和强化学习（RL）。

3.1. Mobius的数据效率显著优于Transformer。

如图2所示，Mobius在训练的所有阶段均取得了显著高于Transformer的MMLU分数。此外，以Transformer在1TB词元时的分数为基线，Mobius仅需0.626× 的数据即可达到相同分数——即Mobius的数据效率是Transformer的1.6×倍。

Mobius实现更高数据效率的机制仍有待充分探索。尽管如此，我们推测在Transformer架构下，各层之间的参数存在相当大的冗

Table 2: Step-aligned comparison between Intern-S2-Mobius-35B and Qwen3.5-35B on a linear-algebra multiple-choice question. Both models select the correct answer (Option C). Token counts are computed using the Qwen3.5-35B tokenizer. Mobius completes the same reasoning steps with fewer tokens, which mainly benefits from the model's elimination of repeated derivation and checks.

Reasoning Trace Comparison

Aligned Step	Intern-S2-Mobius-35B	Qwen3.5-35B
Task framing	**17 tokens.** “The user wants me to evaluate the truth value of two statements regarding linear algebra.”	**20 tokens.** “The user wants me to evaluate two statements and determine their truth values from a list of options.”
Statement 1	**178 tokens.** “Counter-example: Let $v_1 = (1, 0)$ and $v_2 = (2, 0)$. Then $2v_1 - v_2 = 0$, so they are linearly dependent. . . . Statement 1 is False.”	**283 tokens.** “Consider the zero vector. Let $v_1 = (1, 0)$ and $v_2 = (0, 0)$. . . . Consider two parallel vectors. Let $v_1 = (1, 0)$ and $v_2 = (2, 0)$. . . . Statement 1 is **False**.”
Statement 2	**142 tokens.** “A basis is a linearly independent spanning set. Since $\{v_1, \ldots, v_k\}$ spans V and is linearly independent, it forms a basis for V. . . . dim$(V) = k$.”	**197 tokens.** “The set forms a basis for V if it spans V and is linearly independent. . . . Therefore, $\{v_1, \ldots, v_k\}$ is a basis for V. . . . dim$(V) = k$.”
Option matching	**22 tokens.** “Statement 1 is False, Statement 2 is True. This matches Option C.”	**122 tokens.** “The combination is False, True. This corresponds to Option C.”
Repeated derivation and checks	—	**1,147 tokens.** “**Step-by-step derivation:** . . . Counterexample: Let $v_1 = (1, 0)$ and $v_2 = (2, 0)$. . . . **Final check:** . . . If $k = 0$? . . . Works. . . . Option C.”
Visible final answer	**157 tokens.** “The vectors $(1, 0)$ and $(2, 0)$ are linearly dependent because one is a scalar multiple of the other. . . . This is the definition of a basis. . . . ANSWER: C”	**595 tokens.** “**Step 1: Evaluate Statement 1** . . . $2v_1 - 1v_2 = (0, 0)$. . . . Statement 1 is **False**. **Step 2: Evaluate Statement 2** . . . dim(V) is equal to k. . . . ANSWER: C”
Total	**516 tokens**	**2,364 tokens**

3. How about Mobius' performance?

To validate the performance of Mobius, we conducted both training-from-scratch (TFS) and continual pre-training (CPT) experiments. For the TFS experiments, we trained 7B-A1B MoE models — one with Mobius and one with Transformer — on 1TB tokens, and compared their MMLU scores. For the CPT experiments, we used Qwen3.5-35B-A3B as the starting checkpoint, continued pre-training on 1TB tokens, and subsequently performed supervised fine-tuning (SFT) and reinforcement learning (RL).

3.1. Mobius delivers markedly better data efficiency than Transformer.

As shown in Figure 2, Mobius achieves significantly higher MMLU scores than Transformer across all stages of training. Moreover, when using Transformer's score at 1TB tokens as the baseline, Mobius attains the same score with only 0.626× data — that is, Mobius exhibits 1.6× data efficiency of Transformer.

The mechanism by which Mobius achieves higher data efficiency remains to be fully explored. Nevertheless, we hypothesize that under the Transformer architecture, parameters across layers exhibit considerable redun-

余性——例如，同一知识片段可能被冗余存储于多个层中。在转向知识与推理解耦的Mobius架构后，模型能够获得更优的压缩率，从而使其能够以显著更少的训练数据获取足够的知识。

3.2. Mobius以远高于Transformer的推理效率达到Transformer级推理水平。

鉴于从头开始预训练的成本过高，为在大规模上验证Mobius的能力，我们选择从现有的开源基础模型继续训练。如表1所示，从Qwen3.5出发并切换至Mobius架构进行继续预训练，不仅保持了模型的整体能力，而且实际上还增强了其整体能力。

更值得注意的是，切换至Mobius不仅保持了基本的推理能力，还提升了端到端推理效率。如图3所示，在多个评估基准上的汇总结果显示，Mobius的请求吞吐量显著高于Transformer。我们还考察了两种模型的推理长度，如图4所示，该图揭示Mobius端到端推理效率的提升主要源于更短的思维链长度。面对相同的问题时，Mobius以明显更短的推理链解决问题，而Transformer则需要更长的思维链进行深思熟虑。

尽管Mobius较短思维链推理背后的精确机制尚未完全确立，我们推测这可能归因于Mobius固有的潜在推理特性，该特性将深思熟虑过程内化于模型之中，并通过连续可微优化实现更高效的推理。

然而，Mobius究竟是仅仅破解了问题，还是真正提供了一种更高效的推理模式，仍有待确定。通过比较多个案例，我们发现Mobius中缩短的思维链源于后者。表2中的案例来自MMLU-Pro数学，测试了两个基本的线性代数命题。第一个命题是错误的，因为 $\mathbb{R}^2$ 中的两个向量不必线性无关。第二个命题是正确的，因为张成 V 的线性无关集是 V 的一个基，因此其 k 个向量意味着 $\dim(V) = k$。因此正确答案是选项C：错误，正确。

4. Mobius与主流研究的关系

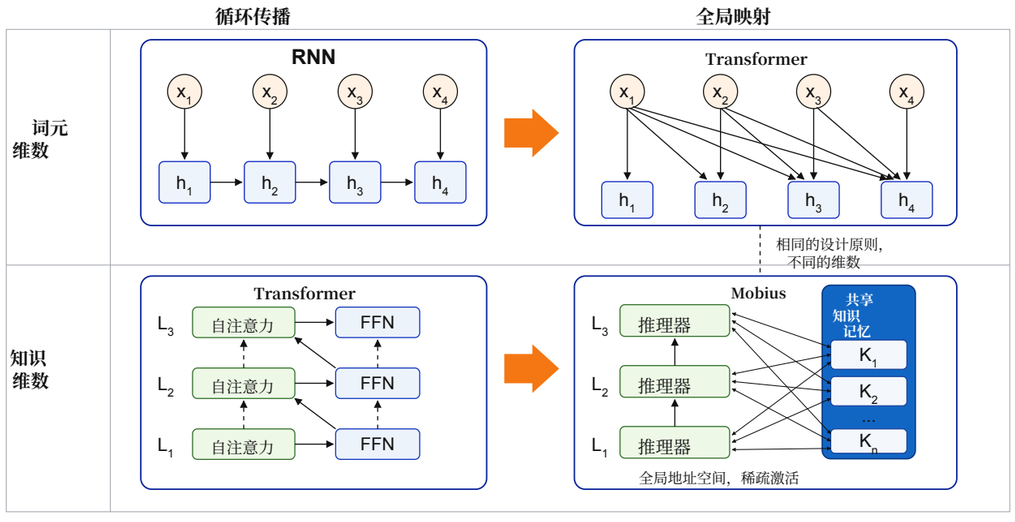


图5：循环神经网络、Transformer与Mobius之间的比较。

dancy—for instance, the same piece of knowledge may be redundantly stored in multiple layers. Upon switching to the knowledge-reasoning-decoupled Mobius architecture, the model can attain superior compression rates, thereby enabling it to acquire sufficient knowledge with substantially less training data.

3.2. Mobius matches Transformer-level reasoning with far greater inference efficiency.

Given the prohibitive cost of pre-training from scratch, to validate Mobius's capabilities at scale, we chose to continue training from an existing open-source base model. As shown in Table 1, starting from Qwen3.5 and switching to the Mobius architecture for continued pre-training not only preserves but actually enhances the model's overall capabilities.

More notably, switching to Mobius not only preserves fundamental reasoning capabilities but also improves end-to-end inference efficiency. As shown in Figure 3, aggregated across multiple evaluation benchmarks, Mobius achieves substantially higher request throughput than Transformer. We also examined the inference length of both models, visualized in Figure 4, which reveals that the improvement in Mobius's end-to-end inference efficiency primarily stems from shorter CoT lengths. When presented with identical problems, Mobius resolves them with markedly shorter reasoning chains, whereas Transformer requires longer CoT for deliberation.

Although the precise mechanism behind Mobius's shorter CoT reasoning remains not fully established, we hypothesize that this may be attributed to Mobius's native latent reasoning characteristic, which internalizes the deliberation process within the model and achieves more efficient reasoning through continuously differentiable optimization.

However, whether Mobius merely hacks the problem or genuinely provides a more efficient mode of reasoning remains to be determined. By comparing multiple cases, we find that the shortened CoT in Mobius stems from the latter. The case in Table 2 is from MMLU-Pro Mathematics and tests two basic linear-algebra statements. The first statement is false because two vectors in $\mathbb{R}^2$ need not be linearly independent. The second statement is true because a linearly independent set that spans V is a basis of V, so its k vectors imply $\dim(V) = k$. The correct answer is therefore Option C: False, True.

4. Mobius' relationship with mainstream research

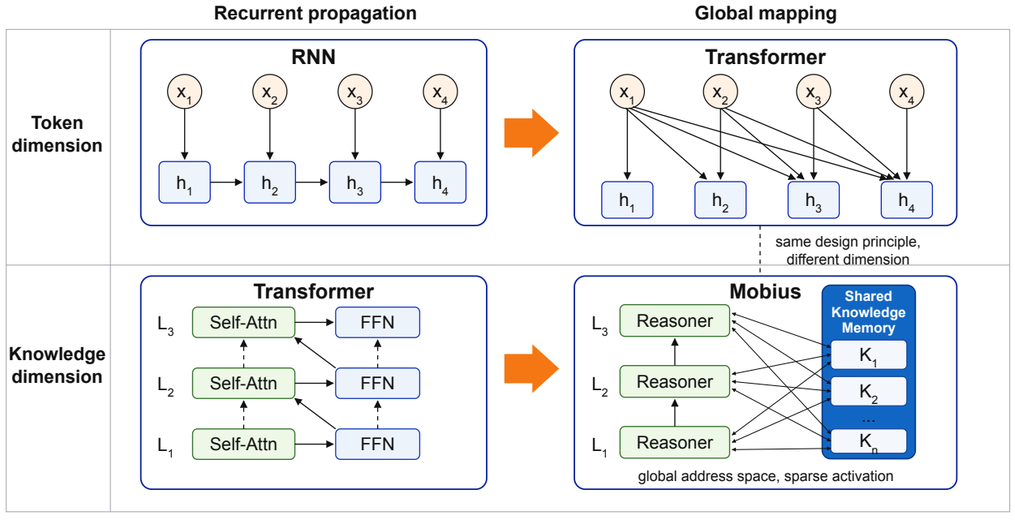


Figure 5: The comparison between RNN, Transformer, and Mobius.

4.1. 潜在推理

思维链（CoT）将推理外部化为中间词元 [71], , 但会产生顺序解码开销。潜在推理则改为在连续状态中进行计算，探索了三个方向：连续思维、循环计算和并行细化。

4.1.1. 连续思维

连续思维方法用可微表示替代离散推理词元，并在推理步骤之间传递这些表示。COCONUT直接将最终隐藏状态作为下一输入嵌入反馈，从而无需将每个状态解码为语言即可进行中间计算 [30]。CODI通过自蒸馏将显式思维链监督压缩为连续状态 [64], , 而SoftCoT则借助轻量级辅助模型生成实例特定的软思维，并将其投影到目标语言模型的表示空间中 [76]。这些方法表明，连续状态能够在缩短或消除显式理由的同时保留有用的推理信息。

这些方法暴露了一小组潜在状态，后续词元可以关注这些状态并将其作为共享上下文重用。Mobius进一步将这种模式打造为原生架构能力：其循环状态针对共享记忆进行精炼，并共同支持多个未来词元，从而在无需冗长轨迹的情况下实现更高密度的推理。

4.1.2. 循环语言模型

循环语言模型通过重复应用共享计算模块来增加有效深度。通用Transformer引入了跨深度参数共享的循环精化 [12], , 随后的分析确立了循环Transformer的计算表达能力 [21]。更近期的工作将有效深度直接与推理联系起来：循环模型可以模拟多步潜在计算，并在推理任务上接近深度大得多的非循环模型 [60], , 而循环深度预训练通过增加潜在迭代次数来实现测试时计算扩展，无需生成额外的推理词元 [19]。

Mobius也通过潜在循环来扩展计算，但每次更新仅跨越少数层，同时保持对完整共享记忆的访问。这种更高的更新频率使得迭代精化更加充分，并将推理集中到信息密度更高的潜在状态中。

4.1.3. 额外潜在计算步骤

另一条研究路线通过引入生成前的额外潜在计算步骤来提升推理能力。暂停令牌方法添加学习得到的占位符以延迟预测并创造额外的计算空间 [24]。隐藏解码在单次前向传播中将每个词元扩展为多个潜在流，利用中间键-值状态作为可复用的计算轨迹 [46]。扩散语言模型通过对连续或掩码表示的迭代精化实现类似效果 [43, 55]。这些方法将计算从显式思维链轨迹转移到潜在轨迹中。

Mobius通过循环潜在细化扩展了这一思想，无需显式占位符或扩散步骤。其推理器反复从共享知识向量记忆中检索，细化高密度潜在状态，并并行解码多个词元。

4.2. 高效推理

从历史上看，语言模型推理的主导范式依赖于单一强模型的自回归单令牌生成，并辅以长思维链。为提升端到端推理效率，两种方法日益受到关注：推测解码（增强推理并行性）与简洁思维链（缩短推理链的序列长度）。

4.1. Latent Reasoning

Chain-of-thought (CoT) externalizes reasoning into intermediate tokens [71], but incurs sequential decoding overhead. Latent reasoning instead performs computation in continuous states, exploring three directions: continuous thought, looped computation, and parallel refinement.

4.1.1. Continuous Thought

Continuous-thought approaches replace discrete reasoning tokens with differentiable representations passed between reasoning steps. COCONUT directly feeds the final hidden state back as the next input embedding, allowing intermediate computation to proceed without decoding each state into language [30]. CODI compresses explicit CoT supervision into continuous states through self-distillation [64], while SoftCoT generates instance-specific soft thoughts with a lightweight assistant and projects them into the representation space of a target language model [76]. These approaches demonstrate that continuous states can preserve useful reasoning information while shortening or eliminating explicit rationales.

These approaches expose a small set of latent states that subsequent tokens can attend to and reuse as shared context. Mobius further makes this pattern a native architectural capability: its recurrent states are refined against the shared Memory and jointly support multiple future tokens, yielding higher-density reasoning without verbose traces.

4.1.2. Looped Language Models

Looped language models increase effective depth by repeatedly applying shared computation blocks. Universal Transformers introduced recurrent refinement across depth with parameter sharing [12], while subsequent analyses established the computational expressivity of Looped Transformers [21]. More recent work connects effective depth directly to reasoning: looped models can emulate multi-step latent computation and approach much deeper non-looped models on reasoning tasks [60], while recurrent-depth pre-training enables test-time compute scaling by increasing the number of latent iterations without generating additional reasoning tokens [19].

Mobius also scales computation through latent recurrence, but performs each update over only a few layers while retaining access to the full shared Memory. This higher update frequency enables more iterative refinement and concentrates reasoning into higher-information-density latent states.

4.1.3. Additional Latent Computation Steps

Another line of work increases reasoning capacity by introducing extra latent computation steps before generation. Pause-token methods add learned placeholders to defer prediction and create additional computation space [24]. Hidden Decoding expands each token into multiple latent streams within a single forward pass, using intermediate key–value states as reusable computation traces [46]. Diffusion language models achieve similar effects through iterative refinement of continuous or masked representations [43, 55]. These methods shift computation from explicit CoT traces to latent trajectories.

Mobius extends this idea through recurrent latent refinement without explicit placeholders or diffusion steps. Its Reasoners repeatedly retrieve from a shared knowledge-vector Memory, refine high-density latent states, and decode multiple tokens in parallel.

4.2. Efficient Reasoning

Historically, the predominant paradigm for language model inference relied on autoregressive single-token generation from a single strong model, augmented with long chains of thought. To improve end-to-end inference efficiency, two approaches have gained increasing traction: Speculative Decoding, which enhances inference parallelism, and Concise CoT, which reduces the sequential length of reasoning chains.

4.2.1. 推测解码

推测解码通过使用廉价的草稿模型提出未来词元的块或树，并与目标模型 [8, 42, 53, 68] 并行验证这些候选词元，从而加速自回归生成。借助拒绝采样校正，该验证保持了目标模型的输出分布。经典方法采用轻量级外部草稿模型，后续研究则基于辅助解码头或预测的隐藏特征 [2, 7, 44] 开发了内部草稿模型。MTP 正是一种内部草稿机制，而非解码协议本身：共享目标主干网络的辅助头预测多个未来词元 [22, 47, 51]。

无论是采用草稿模型还是其他策略，推测解码在词元预测之前仅对全部知识进行一次遍历。相比之下，Mobius在内部执行多轮知识遍历和多词元迭代，在最终解码之前生成信息密度更高的隐藏状态，并原生支持多词元预测。

4.2.2. 简洁思维链

长思维链推理可以提升具有挑战性任务上的表现，但其自回归生成会带来可观的推理成本和延迟。因此，近期工作通过若干互补方法寻求简洁思维链：监督方法从剪枝或配对的长短轨迹中学习，或将显式理由内化为隐式或稠密表示 [9, 13, 14, 36, 73, 79]；基于强化学习的方法通过奖励设计来塑造推理长度，包括可控压缩和长度感知目标 [11, 18, 50, 65]。尽管强化学习能够激发自我纠正、验证和回溯等高级行为 [26, 62], , 仅结果奖励往往偏向更长的轨迹，因为额外的词元能够实现进一步的探索和错误修正 [77]。因此，核心挑战在于降低思维链冗长性，同时不削弱实现准确且稳健推理所需的深思熟虑。

传统的长度正则化可能通过削减深思熟虑来缩短思维链，从而可能损害准确性和鲁棒性。相比之下，Mobius通过多轮潜在思维迭代，生成了更为精炼且更短的响应。

4.3. 架构设计

Mobius通过将知识存储与推理计算解耦，重新设计了Transformer架构的内部信息流。在本节中，我们讨论了其与两大架构方向的关联：注意力机制和残差连接。

4.3.1. 注意力机制

自注意力是Transformer架构的核心组件，通过词元间的成对交互使模型能够捕获全局依赖关系 [69]。然而，全注意力的二次复杂度已成为将Transformer模型扩展到长上下文场景的主要瓶颈。为解决这一问题，大量研究探索了更高效的注意力机制。稀疏注意力通过限制注意力模式来减少不必要的词元交互，而线性注意力则通过替代核化或分解策略重新表述注意力计算以降低复杂度。最近，状态空间模型和门控状态空间变体（如Mamba和门控Delta网络）用循环状态更新取代了显式的注意力计算，为长序列建模提供了高效替代方案 [25, 78]。这些方法主要通过修改计算形式或降低信息交互成本来提高效率。

我们并非直接降低注意力复杂度，而是旨在提升每次注意力操作所提取的信息密度与价值。具体而言，通过将知识与推理解耦并构建全局共享知识库，Mobius虽然会产生更高的检索成本，但为注意力提供了更灵活的选择空间和更高质量的输入，从而以更紧凑的计算轨迹完成推理。

4.2.1. Speculative Decoding

Speculative decoding accelerates autoregressive generation by using a cheap drafter to propose a block, or a tree, of future tokens and verifying these candidates in parallel with the target model [8, 42, 53, 68]. With rejection-sampling correction, this verification preserves the target model's output distribution. While classical methods employ a lightweight external draft model, later work develops internal drafters based on auxiliary decoding heads or predicted hidden features [2, 7, 44]. MTP is such an internal drafting mechanism rather than a decoding protocol itself: auxiliary heads sharing the target backbone predict multiple future tokens [22, 47, 51].

Whether employing a draft model or other strategies, Speculative Decoding traverses all knowledge only once before token prediction. In contrast, Mobius performs multiple rounds of knowledge traversal and multi-token iteration internally, yielding hidden states of higher information density prior to final decoding, with native support for multi-token prediction.

4.2.2. Concise Chain of Thought

Long chain-of-thought (CoT) reasoning can improve performance on challenging tasks, but its autoregressive generation incurs substantial inference cost and latency. Recent work therefore seeks concise CoT through several complementary approaches: supervised methods learn from pruned or paired long–short trajectories, or internalize explicit rationales into implicit or dense representations [9, 13, 14, 36, 73, 79]; RL-based methods shape reasoning length through reward design, including controllable compression and length-aware objectives [11, 18, 50, 65]. Although RL can elicit advanced behaviors such as self-correction, verification, and backtracking [26, 62], outcome-only rewards often favor longer trajectories because additional tokens enable further exploration and error correction [77]. The central challenge is thus to reduce CoT verbosity without removing the deliberation needed for accurate and robust reasoning.

Conventional length regularization may shorten CoT by curtailing deliberation, potentially harming accuracy and robustness. In contrast, Mobius yields far more refined and shorter responses through several rounds of latent thought iteration.

4.3. Architecture Design

Mobius redesigns the internal information flow of Transformer architectures by decoupling knowledge storage from reasoning computation. In this section, we discuss its relationship with two major architectural directions: attention mechanisms and residual connections.

4.3.1. Attention Mechanism

Self-attention is the core component of Transformer architectures, enabling models to capture global dependencies through pairwise interactions among tokens [69]. However, the quadratic complexity of full attention has become a major bottleneck for scaling Transformer models to long-context scenarios. To address this issue, extensive studies have explored more efficient attention mechanisms. Sparse attention reduces unnecessary token interactions by restricting attention patterns, while linear attention reformulates the attention computation to reduce complexity through alternative kernelization or factorization strategies. More recently, state-space models and gated state-space variants, such as Mamba and Gated Delta Networks, replace explicit attention computation with recurrent state updates, providing efficient alternatives for long-sequence modeling [25, 78]. These approaches mainly improve efficiency by modifying the computation form or reducing the cost of information interaction.

Rather than reducing attention complexity directly, we aim to increase the information density and value extracted per attention operation. Specifically, by decoupling knowledge from reasoning and constructing a globally shared knowledge repository, Mobius incur higher retrieval costs but provide attention with a more flexible selection space and higher-quality inputs, thereby completing reasoning with a more compact computation trajectory.

4.3.2. 残差连接

残差连接是扩展深度神经网络的基础组件，通过捷径路径实现稳定优化和有效的信息传播 [31]。后续研究探索了更灵活的连接模式，以改善跨层的信息传输。高速公路网络引入可学习门控来自适应地控制信息流 [66], , 而DenseNet通过允许每一层直接访问先前层的表示来增加连接密度 [35]。最近，超连接和注意力残差进一步研究了如何通过增加连接容量或动态聚合先前层表示来增强残差路径 [39, 80]。尽管有这些改进，现有的残差增强方法主要侧重于强化前向信息传播，即信息从较早层流向较晚层。

Mobius将残差设计从前向信息传播扩展到双向知识访问，而非显式扩展残差通路。由于所有推理阶段均可访问同一知识库，更深层的推理器能够检索其局部层级之外的知识，从而在保持高效计算结构的同时实现更灵活的信息传输。

5. Mobius在若干重点主题上的潜力

5.1. 自我进化

尽管模型和智能体的能力日益增强，但如何使其能够持续从外部来源吸收新知识和新技能仍是一个悬而未决的问题 [40, 56, 59, 72]。

我们的立场是，当前的Transformer架构并不满足自我进化的前提条件。由于知识与推理在Transformer中紧密耦合，此类模型只能通过端到端训练来获取新技能，这不可避免地会同时改变知识与推理能力，从而导致对先前所学技能的灾难性遗忘。适合自我进化的架构应具备一定程度的知识与推理解耦特性，因为这将支持知识存储的无界且高效的扩展，并使模型的推理能力能够跨多个领域泛化。我们提出的Mobius架构具备初步的知识与推理分离特性，展现出相较于Transformer更大的持续学习潜力。

然而，Mobius是否能在真实世界的自我进化场景中表现更优，仍需要基础模型、智能体系统与环境反馈的联合设计，并留待未来工作中加以验证。

5.2. 世界模型

世界模型正成为继语言模型之后的下一个前沿方向，然而如何使模型能够对连续空间输入进行理解与推理，仍是一个悬而未决的问题 [6, 27–29, 33, 61]。

我们的立场是，当前的Transformer架构并不满足世界模型的前提条件。主要基于Transformer构建的模型已被证明能够在太字节参数规模下对离散模态（如语言）进行建模，然而，要利用Transformer对连续空间进行完美建模，可能需要的参数规模将达到拍字节量级。太字节规模语言模型的推理成本已经使现有硬件不堪重负；而拍字节规模世界模型的推理成本将是天文数字。我们提出的Mobius架构原生支持潜在推理，为连续空间建模提供了更强的先验，展现出比Transformer更大的潜力。

尽管如此，Mobius引入的潜在推理先验可能还远不够充分；如何更好地重新设计注意力机制或许更为关键。

5.3. 科学发现

当代人工智能在代码和数学方面已取得显著能力，近期的科学基础模型擅长程序性问题求解 [5, 58]。然而，使模型能够提出颠覆性的

4.3.2. Residual Connection

Residual connections are fundamental components for scaling deep neural networks, enabling stable optimization and effective information propagation through shortcut pathways [31]. Subsequent studies have explored more flexible connection patterns to improve information transmission across layers. Highway Networks introduce learnable gates to adaptively control information flow [66], while DenseNet increases connection density by allowing each layer to directly access previous representations [35]. More recently, Hyper-Connections and Attention Residuals further investigate how to enhance residual pathways by increasing connection capacity or dynamically aggregating previous-layer representations [39, 80]. Despite these improvements, existing residual enhancement methods mainly focus on strengthening forward information propagation, where information flows from earlier layers to later layers.

Mobius extends residual design from forward information propagation toward bidirectional knowledge access, instead of explicitly expanding residual pathways. Since all reasoning stages can access the same knowledge repository, deeper Reasoners can retrieve knowledge beyond their local layer hierarchy, enabling more flexible information transmission while maintaining an efficient computation structure.

5. Mobius' potential on several highlight topics

5.1. Self-Evolving

Although models and agents have grown increasingly capable, how to enable them to continuously absorb new knowledge and skills from external sources remains an open question [40, 56, 59, 72].

Our position is that, **the current Transformer architecture does not satisfy the prerequisites for self-evolution.** Because knowledge and reasoning are tightly coupled in Transformer, such models can only acquire new skills through end-to-end training, which inevitably alters both knowledge and reasoning capabilities simultaneously, thereby causing catastrophic forgetting of previously learned skills. An architecture suitable for self-evolution should exhibit a certain degree of knowledge-reasoning decoupling, as this would support unbounded and efficient expansion of knowledge storage, and enable the model's reasoning capabilities to generalize across multiple domains. The Mobius architecture we propose possesses preliminary knowledge-reasoning separation characteristics, demonstrating greater potential for continual learning compared to Transformer.

Nevertheless, whether Mobius can perform better in real-world self-evolution scenarios still requires the joint design of foundation models, agentic systems, and environmental feedback, and remains to be validated in future work.

5.2. World Model

World models are emerging as the next frontier beyond language models, yet how to enable models to perform understanding and reasoning over continuous-space inputs remains an open question [6, 27–29, 33, 61].

Our position is that, **the current Transformer architecture does not satisfy the prerequisites for world models.** Models built primarily upon Transformer have been demonstrated to model discrete modalities (such as language) at the terabyte parameter scale, yet perfect modeling of continuous space with Transformer would likely require parameter scales on the order of petabytes. The inference cost of terabyte-scale language models already strains existing hardware to its limits; the inference cost of a petabyte-scale world model would be astronomical. The Mobius architecture we propose natively supports latent reasoning, offering a stronger prior for continuous-space modeling and demonstrating greater potential than Transformer.

Nevertheless, the latent reasoning prior introduced by Mobius may prove far from sufficient; how to better reengineer the attention mechanism may be of even greater criticality.

5.3. Scientific Discovery

Contemporary AI has achieved remarkable proficiency in code and mathematics, and recent scientific foundation models excel at procedural problem-solving [5, 58]. Nevertheless, enabling models to formulate disruptive

科学假设和想法仍然是一个未解决的挑战 [23, 48, 70]。

我们的立场是，当前Transformer架构与长思维链相结合的模式并不满足科学发现的先决条件。在现有模式下，模型擅长程序性问题求解；然而，科学发现不仅需要程序性能力，还需要科学直觉以及知识的组合与泛化。Mobius架构从两个方面解决了这一问题：一方面，它将深思熟虑内化到连续可微潜空间 [30, 64]，中，使模型有机会发展出更强大的直觉；另一方面，它扁平化了模型内的所有知识，使更多知识能够同时交互，从而大幅提升跨知识领域组合泛化的可能性。

然而，Mobius是否能在真实科学发现场景中表现更优，可能需要在数据、基础设施、训练算法和优化算法上进行联合优化，并仍有待在未来的工作中加以验证。

5.4. 软硬件协同设计

模型正被扩展到越来越大的参数规模，这表明更大的规模确实能带来涌现智能，同时也将物理硬件的极限推至临界 [6, 33, 37, 41, 57]。

我们的立场是，当前Transformer架构与现有硬件系统的组合并不满足进一步扩展的前提条件。这是因为Transformer架构将知识与推理混杂在同一参数集中，导致部署时几乎需要将所有参数加载到GPU内存中。我们提出的Mobius架构展现出知识与推理的分离，从而开启了在GPU内存中仅稳定保留推理专用参数、而将知识参数主要存储于SSD、并按需将高优先级知识检索并加载至内存的可能性。

从长远来看，Mobius 比 Transformer 更适合软硬件协同设计。然而，要实现这一愿景仍需付出大量努力。

scientific hypotheses and ideas remains an open challenge [23, 48, 70].

Our position is that, **the current paradigm of Transformer architecture combined with long chains of thought does not satisfy the prerequisites for scientific discovery.** Under the present paradigm, models excel at procedural problem-solving; however, scientific discovery demands not only procedural competence but also scientific intuition and the composition and generalization of knowledge. The Mobius architecture addresses this on two fronts: on one hand, it internalizes deliberation into a continuously differentiable latent space [30, 64], affording the model the opportunity to develop more powerful intuition; on the other hand, it flattens all knowledge within the model, enabling a greater volume of knowledge to interact simultaneously and substantially increasing the possibility of compositional generalization across knowledge domains.

Nevertheless, whether Mobius can perform better in real-world scientific discovery scenarios likely requires the joint optimization of data, infrastructure, training algorithms, and optimization algorithms, and remains to be validated in future work.

5.4. Hardware-Software Co-Design

Models are being scaled to ever-larger parameter counts, demonstrating that greater scale does yield emergent intelligence, while simultaneously pushing the limits of physical hardware [6, 33, 37, 41, 57].

Our position is that, **the current combination of Transformer architecture and existing hardware systems does not satisfy the prerequisites for further scaling.** This is because the Transformer architecture commingles knowledge and reasoning within the same parameter set, necessitating the loading of nearly all parameters into GPU memory at deployment. The Mobius architecture we propose exhibits knowledge-reasoning separation, opening the possibility of stably retaining only reasoning-dedicated parameters in GPU memory while storing knowledge parameters predominantly on SSD, with high-priority knowledge retrieved and loaded into memory on demand.

In the long term, Mobius is more amenable to hardware-software co-design than Transformer. However, substantial effort remains before this vision can be realized.

参考文献

[1] Zeyuan Allen-Zhu. ICML 2024 教程：语言模型物理学，2024年7月。项目页面： https://physics.allen-zhu.com/。C[2] Zachary Ankner, Rishab Parthasarathy, Aniruddha Nrusimha, Christopher Rinard, Jonathan Ragan-Kelley, and William Brandon. Hydra：用于 Medusa 解码的序列依赖草稿头。arXiv 预印本 arXiv:2402.05109, 2024. 4.2.1[3] Simran Arora, Sabri Eyuboglu, Michael Zhang, Aman Timalsina, Silas Alberti, Dylan Zinsley, James Zou, Atri Rudra, and Christopher Ré. 简单线性注意力语言模型平衡召回率-吞吐量权衡，2025. 1[4] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. 时空注意力是否足以满足视频理解？在 ICML，2021. 1[5] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. 基于大语言模型的自主化学研究。自然，624:570–578, 2023. 5.3[6] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 语言模型是少样本学习者。在 NeurIPS，2020. 5.2, 5.4[7] Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa：具有多解码头的大语言模型推理加速框架，2024. 4.2.1[8] Charlie Chen, Sebastian Borgeaud, Geoffrey Irving, Jean-Baptiste Lespiau, Laurent Sifre, and John Jumper. 利用推测采样加速大语言模型解码，2023. 4.2.1[9] Jeffrey Cheng and Benjamin Van Durme. 压缩思维链：通过稠密表示实现高效推理。arXiv 预印本 arXiv:2412.13171, 2024. 2.2, 4.2.2[10] Krzysztof Choromanski, Valerii Likhosherstov, Xingyou Song, Richard Davis, Kyle Sarlos, and Adrian Weller. 利用 Performer 重新思考注意力。在 ICLR，2021. 1[11] Muzhi Dai, Shixuan Liu, and Qingyi Si. 面向高效推理的稳定强化学习，2025. 4.2.2[12] Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Lukasz Kaiser. 通用 Transformer。在 ICLR。OpenReview.net, 2019. 4.1.2[13] Yuntian Deng, Yejin Choi, and Stuart Shieber. 从显式思维链到隐式思维链：逐步学习内化思维链。arXiv 预印本 arXiv:2405.14838, 2024. 4.2.2[14] Yuntian Deng, Kiran Prasad, Roland Fernandez, Paul Smolensky, Vishrav Chaudhary, and Stuart Shieber. 通过知识蒸馏实现隐式思维链推理，2023. 4.2.2[15] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT：用于语言理解的深度双向 Transformer 预训练。在 Jill Burstein, Christy Doran, and Thamar Solorio 编辑的《2019年北美计算语言学协会会议论文集：人类语言技术，第1卷（长文和短文）》，第4171–4186页，明尼阿波利斯，明尼苏达州，2019年6月。计算语言学协会。1[16] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 一张图像相当于16x16个词：大规模图像识别 Transformer，2021. 1

References

[1] Zeyuan Allen-Zhu. ICML 2024 Tutorial: Physics of Language Models, July 2024. Project page: https://physics.allen-zhu.com/. C

[2] Zachary Ankner, Rishab Parthasarathy, Aniruddha Nrusimha, Christopher Rinard, Jonathan Ragan-Kelley, and William Brandon. Hydra: Sequentially-dependent draft heads for medusa decoding. *arXiv preprint arXiv:2402.05109*, 2024. 4.2.1

[3] Simran Arora, Sabri Eyuboglu, Michael Zhang, Aman Timalsina, Silas Alberti, Dylan Zinsley, James Zou, Atri Rudra, and Christopher Ré. Simple linear attention language models balance the recall-throughput tradeoff, 2025. 1

[4] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is space-time attention all you need for video understanding? In *ICML*, 2021. 1

[5] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624:570–578, 2023. 5.3

[6] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In *NeurIPS*, 2020. 5.2, 5.4

[7] Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa: Simple llm inference acceleration framework with multiple decoding heads, 2024. 4.2.1

[8] Charlie Chen, Sebastian Borgeaud, Geoffrey Irving, Jean-Baptiste Lespiau, Laurent Sifre, and John Jumper. Accelerating large language model decoding with speculative sampling, 2023. 4.2.1

[9] Jeffrey Cheng and Benjamin Van Durme. Compressed chain of thought: Efficient reasoning through dense representations. *arXiv preprint arXiv:2412.13171*, 2024. 2.2, 4.2.2

[10] Krzysztof Choromanski, Valerii Likhosherstov, Xingyou Song, Richard Davis, Kyle Sarlos, and Adrian Weller. Rethinking attention with performers. In *ICLR*, 2021. 1

[11] Muzhi Dai, Shixuan Liu, and Qingyi Si. Stable reinforcement learning for efficient reasoning, 2025. 4.2.2

[12] Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Lukasz Kaiser. Universal transformers. In *ICLR*. OpenReview.net, 2019. 4.1.2

[13] Yuntian Deng, Yejin Choi, and Stuart Shieber. From explicit cot to implicit cot: Learning to internalize cot step by step. *arXiv preprint arXiv:2405.14838*, 2024. 4.2.2

[14] Yuntian Deng, Kiran Prasad, Roland Fernandez, Paul Smolensky, Vishrav Chaudhary, and Stuart Shieber. Implicit chain of thought reasoning via knowledge distillation, 2023. 4.2.2

[15] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In Jill Burstein, Christy Doran, and Thamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. 1

[16] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale, 2021. 1

[17] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. Transformer电路的一种数学框架。Transformer Circuits Thread, 2021. https://transformer-circuits.pub/2021/framework/index.html. 2

[18] Mehdi Fatemi, Banafsheh Rafiee, Mingjie Tang, and Kartik Talamadupula. 通过强化学习进行简洁推理，2025. 4.2.2

[19] Jonas Geiping、Sean McLeish、Neel Jain、John Kirchenbauer、Siddharth Singh、Brian Bartoldson、Bhavya Kailkhura、Abhinav Bhatele 和 Tom Goldstein。利用潜在推理扩展测试时计算：一种循环深度方法。载于 D. Belgrave、C. Zhang、H. Lin、R. Pascanu、P. Koniusz、M. Ghassemi 和 N. Chen 主编，《神经信息处理系统进展》，第 38 卷，第 41340–41391 页。柯伦联合公司，2025 年。4.1.2

[20] Mor Geva、Roei Schuster、Jonathan Berant 和 Omer Levy。Transformer 前馈层是键值记忆。载于 Marie-Francine Moens、Xuanjing Huang、Lucia Specia 和 Scott Wen-tau Yih 主编，《2021 年自然语言处理经验方法会议论文集》，第 5484–5495 页，线上及多米尼加共和国蓬塔卡纳，2021 年 11 月。计算语言学协会。2

[21] Angeliki Giannou、Shashank Rajput、Jy yong Sohn、Kangwook Lee、Jason D. Lee和Dimitris Papailiopoulos。循环Transformer作为可编程计算机。载于ICML，第11398–11442页，2023年。2.2、4.1.2。

[22] Fabian Gloeckle、Badr Youbi Idrissi、Baptiste Rozière、David Lopez-Paz和Gabriel Synnaeve。通过多令牌预测实现更好更快的大语言模型。见第41届国际机器学习大会论文集，第15706–15734页，2024年。4.2.1

[23] Juraj Gottweis、Wei-Hung Weng、Alexander Daryin、Tao Tu等人。利用共同科学家加速科学发现。《自然》，655(8122)：487–496，2026年。5.3[24] Sachin Goyal、Ziwei Ji、Ankit Singh Rawat、Aditya Krishna Menon、Sanjiv Kumar和Vaishnavh Nagarajan。三思而后言：使用暂停词元训练语言模型。载于B. Kim、Y. Yue、S. Chaudhuri、K. Fragkiadaki、M. Khan和Y. Sun主编，《国际学习表征会议》，第2024卷，第27896–27923页，2024年。4.1.3[25] Albert Gu和Tri Dao。Mamba：基于选择性状态空间的线性时间序列建模。载于《第一届语言建模会议》，2024年。1，4.3.1

[26] 郭达亚，杨德健，张浩伟，宋俊晓，王培懿，朱启豪，徐润欣，张若愚，马世荣，毕晓，张晓康，于兴凯，吴宇，吴志峰，苟志斌，邵志宏，李卓舒，高子怡，刘爱欣，薛冰，王冰璇，吴博超，冯贝，陆成达，赵成刚，邓承奇，阮翀，戴大麦，陈德利，蒋东杰，李尔航，林方云，戴福聪，罗福莉，郝广博，陈冠廷，李国伟，张浩，徐汉伟，丁宏伟，高华佐，曲辉，李辉，郭建忠，李佳石，陈景昌，袁景阳，涂锦豪，邱俊杰，李俊龙，蔡佳良，倪佳琪，梁健，陈瑾，董凯，胡凯，游凯超，高凯歌，关康，黄可欣，于快，王林，张乐聪，赵亮，王立通，张丽月，徐磊，夏乐怡，张明川，张明华，唐明辉，周明旭，李萌，王苗君，李明明，田宁，黄盼盼，张鹏，王前程，陈沁雨，杜秋实，葛瑞琪，张瑞松，潘瑞哲，王润吉，陈瑞杰，金瑞林，陈如意，卢尚浩，周尚燕，陈山煌，叶圣峰，王诗雨，于水平，周顺峰，潘淑婷，李思思，周爽，吴少卿，云涛，裴天，孙天宇，王婷，曾旺丁，刘文，梁文峰，高文军，于文琴，张文涛，肖伟林，安伟，刘晓东，王潇涵，陈晓康，聂晓涛，程欣，刘欣，谢鑫，刘兴超，杨欣宇，李欣远，苏学成，林旭恒，李新奇，金祥月，沈晓瑾，陈晓莎，孙晓文，王晓翔，宋欣楠，周欣怡，王贤祖，单新霞，李永康，王雅琴，魏宇轩，张杨，徐艳红，李瑶，赵瑶，孙耀峰，王耀辉，于毅，张一超，石一凡，熊一良，何颖，

[17] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. https://transformer-circuits.pub/2021/framework/index.html. 2

[18] Mehdi Fatemi, Banafsheh Rafiee, Mingjie Tang, and Kartik Talamadupula. Concise reasoning via reinforcement learning, 2025. 4.2.2

[19] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. In D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen, editors, *Advances in Neural Information Processing Systems*, volume 38, pages 41340–41391. Curran Associates, Inc., 2025. 4.1.2

[20] Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer feed-forward layers are key-value memories. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih, editors, *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 5484–5495, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. 2

[21] Angeliki Giannou, Shashank Rajput, Jy yong Sohn, Kangwook Lee, Jason D. Lee, and Dimitris Papailiopoulos. Looped transformers as programmable computers. In *ICML*, pages 11398–11442, 2023. 2.2, 4.1.2

[22] Fabian Gloeckle, Badr Youbi Idrissi, Baptiste Rozière, David Lopez-Paz, and Gabriel Synnaeve. Better & faster large language models via multi-token prediction. In *Proceedings of the 41st International Conference on Machine Learning*, pages 15706–15734, 2024. 4.2.1

[23] Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, Tao Tu, et al. Accelerating scientific discovery with co-scientist. *Nature*, 655(8122):487–496, 2026. 5.3

[24] Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh Nagarajan. Think before you speak: Training language models with pause tokens. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, editors, *International Conference on Learning Representations*, volume 2024, pages 27896–27923, 2024. 4.1.3

[25] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *First Conference on Language Modeling*, 2024. 1, 4.3.1

[26] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, Bing Xue, Bingxuan Wang, Bochao Wu, Bei Feng, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chong Ruan, Damai Dai, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Hanwei Xu, Honghui Ding, Huazuo Gao, Hui Qu, Hui Li, Jianzhong Guo, Jiashi Li, Jingchang Chen, Jingyang Yuan, Jinhao Tu, Junjie Qiu, Junlong Li, J. L. Cai, Jiaqi Ni, Jian Liang, Jin Chen, Kai Dong, Kai Hu, Kaichao You, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Liang Zhao, Litong Wang, Liyue Zhang, Lei Xu, Leyi Xia, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingxu Zhou, Meng Li, Miaojun Wang, Mingming Li, Ning Tian, Panpan Huang, Peng Zhang, Qiancheng Wang, Qinyu Chen, Qiushi Du, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, R. J. Chen, R. L. Jin, Ruyi Chen, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shengfeng Ye, Shiyu Wang, Shuiping Yu, Shunfeng Zhou, Shuting Pan, S. S. Li, Shuang Zhou, Shaoqing Wu, Tao Yun, Tian Pei, Tianyu Sun, T. Wang, Wangding Zeng, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, W. L. Xiao, Wei An, Xiaodong Liu, Xiaohan Wang, Xiaokang Chen, Xiaotao Nie, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, X. Q. Li, Xiangyue Jin, Xiaojin Shen, Xiaosha Chen, Xiaowen Sun, Xiaoxiang Wang, Xinnan Song, Xinyi Zhou, Xianzu Wang, Xinxia Shan, Y. K. Li, Y. Q. Wang, Y. X. Wei, Yang Zhang, Yanhong Xu, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Wang, Yi Yu, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He,

朴一石，王宜松，谭宜轩，马一洋，刘一远，郭永强，欧媛，王玉端，龚玥，邹宇恒，何宇佳，熊云帆，罗宇翔，游宇翔，刘宇轩，周宇阳，朱宇轩，黄艳萍，李耀辉，郑毅，朱宇辰，马云仙，唐颖，查宇坤，严宇婷，任泽中，任泽辉，沙张丽，付哲，徐哲安，谢振达，张正彦，郝哲文，马志诚，严志刚，吴志宇，顾子惠，朱子佳，刘子俊，李子林，谢子威，宋子扬，潘子正，黄震，徐志鹏，张忠宇，张震。Deepseek-r1通过强化学习激励大语言模型进行推理。自然，645(8081)：633–638，2025。2，2.2，4.2.2

[27] David Ha和Jürgen Schmidhuber。世界模型。载于《神经信息处理系统进展》，第31卷，2018年。5.2[28] Danijar Hafner、Timothy Lillicrap、Ian Fischer、Ruben Villegas、David Ha、Honglak Lee和James Davidson。学习潜在动力学以从像素进行规划。载于《国际机器学习大会》，第2555–2565页。PMLR，2019年。[29] Danijar Hafner、Jürgen Pasukonis、Jimmy Ba和Timothy Lillicrap。通过世界模型掌握多样领域。arXiv预印本arXiv:2301.04104，2023年。5.2[30] Shibo Hao、Sainbayar Sukhbaatar、DiJia Su、Xian Li、Zhiting Hu、Jason Weston和Yuandong Tian。训练大型语言模型在连续潜在空间中进行推理。arXiv预印本arXiv:2412.06769，2024年。4.1.1，5.3[31] Kaiming He、Xiangyu Zhang、Shaoqing Ren和Jian Sun。用于图像识别的深度残差学习。载于《IEEE计算机视觉与模式识别会议论文集》，第770–778页，2016年。2，2.1，4.3.2[32] Kaiming He、Xiangyu Zhang、Shaoqing Ren和Jian Sun。深度残差网络中的恒等映射。载于《欧洲计算机视觉会议》，第630–645页。施普林格，2016年。2.1[33] Jordan Hoffmann、Sebastian Borgeaud、Arthur Mensch、Elena Buchatskaya、Trevor Cai、Eliza Rutherford、Diego de Las Casas、Lisa Anne Hendricks、Johannes Welbl、Aidan Clark、Tom Hennigan、Eric Noland、Katie Millican、George van den Driessche、Bogdan Damoc、Aurelia Guy、Simon Osindero、Karen Simonyan、Erich Elsen、Oriol Vinyals、Jack W. Rae和Laurent Sifre。训练计算最优的大型语言模型。载于《神经信息处理系统进展》，第35卷，第30016–30030页。柯伦联合公司，2022年。1，1，5.2，5.4[34] Gao Huang、Zhuang Liu、Laurens van der Maaten和Kilian Q. Weinberger。密集连接卷积网络。载于CVPR，第4700–4708页，2017年。2，2.1[35] Gao Huang、Zhuang Liu、Laurens van der Maaten和Kilian Q. Weinberger。密集连接卷积网络。载于《IEEE计算机视觉与模式识别会议论文集》，第4700–4708页，2017年。4.3.2[36] Yu Kang、Xianghui Sun、Liangyu Chen和Wei Zou。C3ot：在不影响有效性的前提下生成更短的思维链。载于《AAAI人工智能会议论文集》，第39卷，第24312–24320页，2025年。4.2.2[37] Jared Kaplan、Sam McCandlish、Tom Henighan、Tom B. Brown、Benjamin Chess、Rewon Child、Scott Gray、Alec Radford、Jeffrey Wu和Dario Amodei。神经语言模型的缩放定律，2020年。1，1，5.4[38] Angelos Katharopoulos、Apoorv Vyas、Nikolaos Pappas和François Fleuret。Transformer是RNN：具有线性注意力的快速自回归Transformer。载于ICML，第5156–5165页，2020年。1[39] Kimi Team。注意力残差。arXiv预印本arXiv:2603.15031，2026年。2.1，4.3.2[40] James Kirkpatrick、Razvan Pascanu、Neil Rabinowitz、Joel Veness、Guillaume Desjardins、Andrei A. Rusu、Kieran Milan、John Quan、Tiago Ramalho、Agnieszka Grabska-Barwinska等人。克服神经网络中的灾难性遗忘。《美国国家科学院院刊》，114(13)：3521–3526，2017年。5.1

Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yuan Ou, Yuduan Wang, Yue Gong, Yuheng Zou, Yujia He, Yunfan Xiong, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Y. X. Zhu, Yanping Huang, Yaohui Li, Yi Zheng, Yuchen Zhu, Yunxian Ma, Ying Tang, Yukun Zha, Yuting Yan, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhicheng Ma, Zhigang Yan, Zhiyu Wu, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Zizheng Pan, Zhen Huang, Zhipeng Xu, Zhongyu Zhang, and Zhen Zhang. Deepseek-r1 incentivizes reasoning in llms through reinforcement learning. *Nature*, 645(8081):633–638, 2025. 2, 2.2, 4.2.2

[27] David Ha and Jürgen Schmidhuber. World models. In *Advances in Neural Information Processing Systems*, volume 31, 2018. 5.2

[28] Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning*, pages 2555–2565. PMLR, 2019.

[29] Danijar Hafner, Jürgen Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023. 5.2

[30] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769*, 2024. 4.1.1, 5.3

[31] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016. 2, 2.1, 4.3.2

[32] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Identity mappings in deep residual networks. In *European conference on computer vision*, pages 630–645. Springer, 2016. 2.1

[33] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Oriol Vinyals, Jack W. Rae, and Laurent Sifre. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 30016–30030. Curran Associates, Inc., 2022. 1, 1, 5.2, 5.4

[34] Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely connected convolutional networks. In *CVPR*, pages 4700–4708, 2017. 2, 2.1

[35] Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 4700–4708, 2017. 4.3.2

[36] Yu Kang, Xianghui Sun, Liangyu Chen, and Wei Zou. C3ot: Generating shorter chain-of-thought without compromising effectiveness. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 24312–24320, 2025. 4.2.2

[37] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. 1, 1, 5.4

[38] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are rnns: Fast autoregressive transformers with linear attention. In *ICML*, pages 5156–5165, 2020. 1

[39] Kimi Team. Attention residuals. *arXiv preprint arXiv:2603.15031*, 2026. 2.1, 4.3.2

[40] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017. 5.1

[41] Woosuk Kwon、Zhuohan Li、Siyuan Zhuang、Ying Sheng、Lianmin Zheng、Cody Hao Yu、Joseph Gonzalez、Hao Zhang和Ion Stoica。基于分页注意力的大语言模型服务高效内存管理。见《第29届操作系统原理研讨会论文集》，第611–626页。ACM，2023年。5.4[42] Yaniv Leviathan、Matan Kalman和Yossi Matias。通过推测解码实现Transformer的快速推理。见《国际机器学习大会》，第19274–19286页。PMLR，2023年。4.2.1[43] Xiang Li、John Thickstun、Ishaan Gulrajani、Percy S Liang和Tatsunori B Hashimoto。Diffusion-lm改进了可控文本生成。见S. Koyejo、S. Mohamed、A. Agarwal、D. Belgrave、K. Cho和A. Oh编，《神经信息处理系统进展》，第35卷，第4328–4343页。柯伦联合公司，2022年。2.2，4.1.3[44] Yuhui Li、Fangyun Wei、Chao Zhang和Hongyang Zhang。Eagle：推测采样需要重新思考特征不确定性。见《第41届国际机器学习大会论文集》，第28935–28948页，2024年。4.2.1[45] Hunter Lightman、Vineet Kosaraju、Yura Burda、Harri Edwards、Bowen Baker、Teddy Lee、Jan Leike、John Schulman、Ilya Sutskever和Karl Cobbe。让我们逐步验证，2023年。1，2.2[46] Aiwei Liu、Cheng Shi、Chuhan Wu、Ci Lei、Di Lu、Donald He、Fan Zhang、Fanhao Kong、Feifei Zhang、Guan Wang等。大规模隐藏解码：面向大语言模型的潜在计算扩展。arXiv预印本arXiv:2607.08186，2026年。4.1.3[47] Aixin Liu、Bei Feng、Bing Xue、Bingxuan Wang、Bochao Wu、Chengda Lu、Chenggang Zhao、Chengqi Deng、Chenyu Zhang、Chong Ruan等。Deepseek-v3技术报告。arXiv预印本arXiv:2412.19437，2024年。4.2.1[48] Chris Lu、Cong Lu、Robert Tjarko Lange、Jakob Foerster、Jeff Clune和David Ha。人工智能科学家：迈向全自动开放式科学发现，2024年。5.3[49] Wenjie Ma、Jingxuan He、Charlie Snell、Tyler Griggs、Sewon Min和Matei Zaharia。推理模型无需思考也能有效，2025年。1[50] Xinyin Ma、Guangnian Wan、Runpeng Yu、Gongfan Fang和Xinchao Wang。Cot-valve：长度可压缩的思维链调优。见《第63届计算语言学协会年会论文集（第1卷：长论文）》，第6025–6035页，2025年。4.2.2[51] Somesh Mehra、Javier Alonso Garcia和Lukas Mauch。面向高效大语言模型推理的多令牌预测，2025年。4.2.1[52] Kevin Meng、David Bau、Alex Andonian和Yonatan Belinkov。定位并编辑GPT中的事实关联。见S. Koyejo、S. Mohamed、A. Agarwal、D. Belgrave、K. Cho和A. Oh编，《神经信息处理系统进展》，第35卷，第17359–17372页。柯伦联合公司，2022年。2[53] Xupeng Miao、Gabriele Oliaro、Zhihao Zhang、Xinhao Cheng、Zeyu Wang、Zhengxin Zhang、Rae Ying Yee Wong、Alan Zhu、Lijie Yang、Xiaoxiang Shi等。Specinfer：基于树状推测推理与验证加速大语言模型服务。见《第29届ACM国际编程语言与操作系统架构支持会议论文集》，第3卷，第932–949页，2024年。4.2.1[54] Niklas Muennighoff、Zitong Yang、Weijia Shi、Xiang Lisa Li、Li Fei-Fei、Hannaneh Hajishirzi、Luke Zettlemoyer、Percy Liang、Emmanuel Candès和Tatsunori Hashimoto。s1：简单的测试时扩展。见Christos Christodoulopoulos、Tanmoy Chakraborty、Carolyn Rose和Violet Peng编，《2025年自然语言处理实证方法会议论文集》，第20275–20321页，中国苏州，2025年11月。计算语言学协会。1[55] Shen Nie、Fengqi Zhu、Zebin You、Xiaolu Zhang、Jingyang Ou、Jun Hu、Jun Zhou、Yankai Lin、Ji-Rong Wen和Chongxuan LI。大语言扩散模型。见D. Belgrave、C. Zhang、H. Lin、R. Pascanu、P. Koniusz、M. Ghassemi和N. Chen编，《神经信息处理系统进展》，第38卷，第50608–50646页。柯伦联合公司，2025年。2.2，4.1.3

[41] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pages 611–626. ACM, 2023. 5.4

[42] Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *International Conference on Machine Learning*, pages 19274–19286. PMLR, 2023. 4.2.1

[43] Xiang Li, John Thickstun, Ishaan Gulrajani, Percy S Liang, and Tatsunori B Hashimoto. Diffusion-lm improves controllable text generation. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems*, volume 35, pages 4328–4343. Curran Associates, Inc., 2022. 2.2, 4.1.3

[44] Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. Eagle: speculative sampling requires rethinking feature uncertainty. In *Proceedings of the 41st International Conference on Machine Learning*, pages 28935–28948, 2024. 4.2.1

[45] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step, 2023. 1, 2.2

[46] Aiwei Liu, Cheng Shi, Chuhan Wu, Ci Lei, Di Lu, Donald He, Fan Zhang, Fanhao Kong, Feifei Zhang, Guan Wang, et al. Hidden decoding at scale: Latent computation scaling for large language models. *arXiv preprint arXiv:2607.08186*, 2026. 4.1.3

[47] Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, et al. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024. 4.2.1

[48] Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The ai scientist: Towards fully automated open-ended scientific discovery, 2024. 5.3

[49] Wenjie Ma, Jingxuan He, Charlie Snell, Tyler Griggs, Sewon Min, and Matei Zaharia. Reasoning models can be effective without thinking, 2025. 1

[50] Xinyin Ma, Guangnian Wan, Runpeng Yu, Gongfan Fang, and Xinchao Wang. Cot-valve: Length-compressible chain-of-thought tuning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6025–6035, 2025. 4.2.2

[51] Somesh Mehra, Javier Alonso Garcia, and Lukas Mauch. On multi-token prediction for efficient llm inference, 2025. 4.2.1

[52] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372. Curran Associates, Inc., 2022. 2

[53] Xupeng Miao, Gabriele Oliaro, Zhihao Zhang, Xinhao Cheng, Zeyu Wang, Zhengxin Zhang, Rae Ying Yee Wong, Alan Zhu, Lijie Yang, Xiaoxiang Shi, et al. Specinfer: Accelerating large language model serving with tree-based speculative inference and verification. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 3*, pages 932–949, 2024. 4.2.1

[54] Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori Hashimoto. s1: Simple test-time scaling. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 20275–20321, Suzhou, China, November 2025. Association for Computational Linguistics. 1

[55] Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan LI. Large language diffusion models. In D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen, editors, *Advances in Neural Information Processing Systems*, volume 38, pages 50608–50646. Curran Associates, Inc., 2025. 2.2, 4.1.3

[56] German I. Parisi、Ronald Kemker、Jose L. Part、Christopher Kanan和Stefan Wermter。基于神经网络的持续终身学习：综述。《Neural Networks》，113:54–71，2019年。5.1[57] Samyam Rajbhandari、Jeff Rasley、Olatunji Ruwase和Yuxiong He。Zero：面向万亿参数模型训练的显存优化。载于SC20：高性能计算、网络、存储与分析国际会议，第1–16页。IEEE，2020年。5.4[58] Bernardino Romera-Paredes、Mohammadamin Barekatain、Alexander Novikov、Matej Balog、M. Pawan Kumar、Emilien Dupont、Francisco J. R. Ruiz、Jordan Ellenberg等。Funsearch：利用大语言模型在数学科学中做出新发现。《自然》，625:468–475，2024年。5.3[59] Adam Santoro、Sergey Bartunov、Matthew M. Botvinick、Daan Wierstra和Timothy P. Lillicrap。基于记忆增强神经网络的单样本学习。CoRR，abs/1605.06065，2016年。5.1[60] Nikunj Saunshi、Nishanth Dikkala、Zhiyuan Li、Sanjiv Kumar和Sashank J. Reddi。基于潜在思维的推理：论循环Transformer的威力。载于Y. Yue、A. Garg、N. Peng、F. Sha和R. Yu编，《国际学习表征会议》，第2025卷，第14855–14881页，2025年。4.1.2[61] Julian Schrittwieser、Ioannis Antonoglou、Thomas Hubert、Karen Simonyan、Laurent Sifre、Simon Schmitt、Arthur Guez、Edward Lockhart、Demis Hassabis、David Silver等。通过习得模型进行规划，掌握雅达利、围棋、国际象棋和将棋。《自然》，588:604–609，2020年。5.2[62] Zhihong Shao、Peiyi Wang、Qihao Zhu、Runxin Xu、Junxiao Song、Xiao Bi、Haowei Zhang、Mingchuan Zhang、Y. K. Li、Y. Wu和Daya Guo。Deepseekmath：推动开放语言模型数学推理的极限，2024年。4.2.2[63] Noam Shazeer、Azalia Mirhoseini、Krzysztof Maziarz、Andy Davis、Quoc Le、Geoffrey Hinton和Jeff Dean。稀疏门控混合专家层。载于ICLR，2017年。2.3[64] Zhenyi Shen、Hanqi Yan、Linhai Zhang、Zhanghao Hu、Yali Du和Yulan He。CODI：通过自蒸馏将思维链压缩至连续空间。载于Christos Christodoulopoulos、Tanmoy Chakraborty、Carolyn Rose和Violet Peng编，《2025年自然语言处理实证方法会议论文集》，第677–693页，中国苏州，2025年11月。计算语言学协会。4.1.1，5.3[65] Mingyang Song和Mao Zheng。先走后跑！基于强化学习的简洁大语言模型推理，2025年。4.2.2[66] Rupesh K. Srivastava、Klaus Greff和Jürgen Schmidhuber。训练极深网络。载于《神经信息处理系统进展》，第2377–2385页，2015年。4.3.2[67] Rupesh K. Srivastava、Klaus Greff和Jürgen Schmidhuber。高速公路网络。载于ICML深度学习研讨会，2015年。2[68] Mitchell Stern、Noam Shazeer和Jakob Uszkoreit。深度自回归模型的分块并行解码。《神经信息处理系统进展》，31，2018年。4.2.1[69] Ashish Vaswani、Noam Shazeer、Niki Parmar、Jakob Uszkoreit、Llion Jones、Aidan N Gomez、Łukasz Kaiser和Illia Polosukhin。注意力就是你所需要的一切。载于I. Guyon、U. Von Luxburg、S. Bengio、H. Wallach、R. Fergus、S. Vishwanathan和R. Garnett编，《神经信息处理系统进展》，第30卷。柯伦联合公司，2017年。1，2，4.3.1[70] Yuru Wang、Lejun Cheng、Yuxin Zuo、Sihang Zeng、Bingxiang He、Che Jiang、Junlin Yang、Yuchong Wang、Kaikai Zhao、Weifeng Huang、Kai Tian、Zhenzhao Yuan、Jincheng Zhong、Weizhi Wang、Ning Ding、Bowen Zhou和Kaiyan Zhang。Naturebench：编码智能体能否达到自然系列论文已发表的SOTA水平？，2026年。5.3[71] Jason Wei、Xuezhi Wang、Dale Schuurmans、Maarten Bosma、Brian Ichter、Fei Xia、Ed H. Chi、Quoc V. Le和Denny Zhou。思维链提示激发大语言模型的推理能力。载于NeurIPS，2022年。1，2，2.2，4.1[72] Jason Weston、Sumit Chopra和Antoine Bordes。记忆网络。载于ICLR，2015年。5.1

[56] German I. Parisi, Ronald Kemker, Jose L. Part, Christopher Kanan, and Stefan Wermter. Continual lifelong learning with neural networks: A review. *Neural Networks*, 113:54–71, 2019. 5.1

[57] Samyam Rajbhandari, Jeff Rasley, Olatunji Ruwase, and Yuxiong He. Zero: Memory optimizations toward training trillion parameter models. In *SC20: International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 1–16. IEEE, 2020. 5.4

[58] Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan Ellenberg, et al. Funsearch: Making new discoveries in mathematical sciences using large language models. *Nature*, 625:468–475, 2024. 5.3

[59] Adam Santoro, Sergey Bartunov, Matthew M. Botvinick, Daan Wierstra, and Timothy P. Lillicrap. One-shot learning with memory-augmented neural networks. *CoRR*, abs/1605.06065, 2016. 5.1

[60] Nikunj Saunshi, Nishanth Dikkala, Zhiyuan Li, Sanjiv Kumar, and Sashank J. Reddi. Reasoning with latent thoughts: On the power of looped transformers. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu, editors, *International Conference on Learning Representations*, volume 2025, pages 14855–14881, 2025. 4.1.2

[61] Julian Schrittwieser, Ioannis Antonoglou, Thomas Hubert, Karen Simonyan, Laurent Sifre, Simon Schmitt, Arthur Guez, Edward Lockhart, Demis Hassabis, David Silver, et al. Mastering atari, go, chess and shogi by planning with a learned model. *Nature*, 588:604–609, 2020. 5.2

[62] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. 4.2.2

[63] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Sparsely-gated mixture-of-experts layers. In *ICLR*, 2017. 2.3

[64] Zhenyi Shen, Hanqi Yan, Linhai Zhang, Zhanghao Hu, Yali Du, and Yulan He. CODI: Compressing chain-of-thought into continuous space via self-distillation. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 677–693, Suzhou, China, November 2025. Association for Computational Linguistics. 4.1.1, 5.3

[65] Mingyang Song and Mao Zheng. Walk before you run! concise llm reasoning via reinforcement learning, 2025. 4.2.2

[66] Rupesh K. Srivastava, Klaus Greff, and Jürgen Schmidhuber. Training very deep networks. In *Advances in Neural Information Processing Systems*, pages 2377–2385, 2015. 4.3.2

[67] Rupesh K. Srivastava, Klaus Greff, and Jürgen Schmidhuber. Highway networks. In *ICML Deep Learning Workshop*, 2015. 2

[68] Mitchell Stern, Noam Shazeer, and Jakob Uszkoreit. Blockwise parallel decoding for deep autoregressive models. *Advances in Neural Information Processing Systems*, 31, 2018. 4.2.1

[69] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. 1, 2, 4.3.1

[70] Yuru Wang, Lejun Cheng, Yuxin Zuo, Sihang Zeng, Bingxiang He, Che Jiang, Junlin Yang, Yuchong Wang, Kaikai Zhao, Weifeng Huang, Kai Tian, Zhenzhao Yuan, Jincheng Zhong, Weizhi Wang, Ning Ding, Bowen Zhou, and Kaiyan Zhang. Naturebench: Can coding agents match the published sota of nature-family papers?, 2026. 5.3

[71] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS*, 2022. 1, 2, 2.2, 4.1

[72] Jason Weston, Sumit Chopra, and Antoine Bordes. Memory networks. In *ICLR*, 2015. 5.1

[73] 夏鹤鸣、梁泽涛、李文杰、李永琪和李文杰。Tokenskip：大语言模型中可控的思维链压缩。载于2025年自然语言处理实证方法会议论文集，第3351–3363页，2025年。4.2.2[74] 谢振达、魏一轩、曹焕奇、赵成刚、邓承奇、李佳石、戴大麦、高华佐、常江、于快、赵亮、周尚燕、徐哲安、张正彦、曾旺丁、胡圣丁、王玉清、袁景阳、王林和梁文峰。mhc：流形约束超连接，2026年。2.1[75] 熊瑞斌、杨云昌、何迪、郑凯、郑淑欣、邢晨、张会帅、兰艳艳、王立威和刘铁岩。论Transformer架构中的层归一化。载于国际机器学习大会，第10524–10533页。PMLR，2020年。2.1[76] 徐一戈、郭旭、曾志伟和苗春燕。SoftCoT：用于大语言模型高效推理的软思维链。载于车万翔、Joyce Nabende、Ekaterina Shutova和Mohammad Taher Pilehvar主编，《第63届计算语言学协会年会论文集》（第一卷：长文），第23336–23351页，奥地利维也纳，2025年7月。计算语言学协会。4.1.1[77] 杨世明、佟宇轩、牛新尧、Graham Neubig和岳翔。揭秘长链思

思维推理。见《国际机器学习大会》，第71177–71209页。PMLR，2025年。4.2.2[78] 杨松林、Jan Kautz和Ali Hatamizadeh。门控Delta网络：利用增量规则改进Mamba2。见《第十三届国际学习表征会议》，2025年。1，4.3.1[79] 于萍、徐静、Jason Weston和Ilia Kulikov。将系统2蒸馏到系统1，2024年。4.2.2[80] 朱德发、黄鸿志、黄子豪、曾宇涛、毛云瑶、吴邦谷、闵启阳和周迅。超连接。见《国际学习表征会议》，2025年。2.1，4.3.2[81] 邹一诚等。Intern-s1-pro：万亿规模科学多模态基础模型，2026年。1

[73] Heming Xia, Chak Tou Leong, Wenjie Wang, Yongqi Li, and Wenjie Li. Tokenskip: Controllable chain-of-thought compression in llms. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 3351–3363, 2025. 4.2.2

[74] Zhenda Xie, Yixuan Wei, Huanqi Cao, Chenggang Zhao, Chengqi Deng, Jiashi Li, Damai Dai, Huazuo Gao, Jiang Chang, Kuai Yu, Liang Zhao, Shangyan Zhou, Zhean Xu, Zhengyan Zhang, Wangding Zeng, Shengding Hu, Yuqing Wang, Jingyang Yuan, Lean Wang, and Wenfeng Liang. mhc: Manifold-constrained hyper-connections, 2026. 2.1

[75] Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tieyan Liu. On layer normalization in the transformer architecture. In *International conference on machine learning*, pages 10524–10533. PMLR, 2020. 2.1

[76] Yige Xu, Xu Guo, Zhiwei Zeng, and Chunyan Miao. SoftCoT: Soft chain-of-thought for efficient reasoning with LLMs. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 23336–23351, Vienna, Austria, July 2025. Association for Computational Linguistics. 4.1.1

[77] Shiming Yang, Yuxuan Tong, Xinyao Niu, Graham Neubig, and Xiang Yue. Demystifying long chain-of-thought reasoning. In *International Conference on Machine Learning*, pages 71177–71209. PMLR, 2025. 4.2.2

[78] Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving mamba2 with delta rule. In *The Thirteenth International Conference on Learning Representations*, 2025. 1, 4.3.1

[79] Ping Yu, Jing Xu, Jason Weston, and Ilia Kulikov. Distilling system 2 into system 1, 2024. 4.2.2

[80] Defa Zhu, Hongzhi Huang, Zihao Huang, Yutao Zeng, Yunyao Mao, Banggu Wu, Qiyang Min, and Xun Zhou. Hyper-connections. In *International Conference on Learning Representations*, 2025. 2.1, 4.3.2

[81] Yicheng Zou et al. Intern-s1-pro: Scientific multimodal foundation model at trillion scale, 2026. 1

A. 作者列表

A.1. 核心贡献者（按贡献排序）

算法设计：

- 架构：Ermo Hua*，Xiangyu Hong*，Che Jiang*，Baiting Wu，Cheng Liang，Youbang Sun，Biqing Qi，Qipeng Guo
- 训练：Baiting Wu*，Chengqi Lv*，Ermo Hua*，Xiangyu Hong*，Weida Wang，Ning Ding，Wenwei Zhang

基础设施：

- 训练：韩静王*，洪翔宇*，顾一诚*，华尔默*，余山，侯浩政，钱建民，侯杰，田忠波，王辉
- 推理：钱瑶*，吴柏廷*，华尔默*，丁继峰，马宁生，吕涵，金敏熙，田忠波，王辉

项目负责人：华尔默

项目顾问：张奇，陈凯，林达华，周博文

A.2. 完整列表（按字符排序）

陈凯，丁继峰，丁宁，葛嘉烨，顾立新，顾一诚，郭启鹏，华尔默，黄海安，侯浩政，侯杰，洪翔宇，蒋哲，金敏熙，梁成，林达华，林天扬，刘大为，刘奎坤，吕承启，吕海军，吕涵，马宁生，齐碧青，钱建民，苏诗雅，孙有邦，唐焕泽，田忠波，韩静王，王瑞，王婷，王毅，吴柏廷，徐俊，杨博文，王辉，王维达，叶浩辰，余嘉硕，余山，余晓艺，曾启瑞，张奇，张明，张文伟，周博文，周欣宇

B. 专家激活模式

我们进一步比较了第3节中介绍的两种训练方案下的专家激活模式。如图6所示，从头训练的Mobius-7B模型（左）在推理层上表现出相对均匀的激活分布。而Intern-S2-Mobius-35B（右）从Qwen3.5-35B持续预训练而来，其专家激活范围更广，同时保留了显著的分块对角模式。后者表明专家路由仍受到源检查点层特定组织的影响。

这些观察结果表明，持续预训练可以拓宽专家利用范围，但无法完全消除架构转换引入的路由先验。因此，从头训练Mobius可能允许更灵活地访问共享专家池；这是否能转化为更强的模型能力，需要在可比规模下进行受控评估。

C. 组合泛化玩具任务

我们从大语言模型物理学 [1]中选取了一个组合泛化任务。在该任务上，我们将Transformer与我们尚未发布的、更新的Mobius架构进行了比较。如图7所示，Mobius在收敛效率和最终分数上均取得了显著优势。该组合泛化任务以500个实体的单跳知识和400个实体的双跳知识作为训练集，以其余100个实体的双跳知识作为测试集。通过比较测试集分数，我们考察模型是否能够真正学习不同知识片段之间的联系，并实现组合泛化。

D. 案例研究：思维链推理

我们进一步在第3节中展示了一个来自生物学领域的示例。在该案例中，Mobius同样生成了比Transformer短得多的推理轨迹。

A. Author List

A.1. Core Contributors (Sorted by Contribution)

Algorithm Design:

- **Architecture:** Ermo Hua*, Xiangyu Hong*, Che Jiang*, Baiting Wu, Cheng Liang, Youbang Sun, Biqing Qi, Qipeng Guo
- **Training:** Baiting Wu*, Chengqi Lv*, Ermo Hua*, Xiangyu Hong*, Weida Wang, Ning Ding, Wenwei Zhang

Infrastructure:

- **Training:** Hanjing Wang*, Xiangyu Hong*, Yicheng Gu*, Ermo Hua*, Shan Yu, Haozheng Hou, Jianmin Qian, Jie Hou, Zhongbo Tian, Hui Wang
- **Inference:** Qian Yao*, Baiting Wu*, Ermo Hua*, Jifeng Ding, Ningsheng Ma, Han Lv, Minxi Jin, Zhongbo Tian, Hui Wang

Project Leader: Ermo Hua

Project Advisors: Qi Zhang, Kai Chen, Dahua Lin, Bowen Zhou

A.2. Full List (Sorted by Character)

Kai Chen, Jifeng Ding, Ning Ding, Jiaye Ge, Lixin Gu, Yicheng Gu, Qipeng Guo, Ermo Hua, Haian Huang, Haozheng Hou, Jie Hou, Xiangyu Hong, Che Jiang, Minxi Jin, Cheng Liang, Dahua Lin, Tianyang Lin, Dawei Liu, Kuikun Liu, Chengqi Lv, Haijun Lv, Han Lv, Ningsheng Ma, Biqing Qi, Jianmin Qian, Shiya Su, Youbang Sun, Huanze Tang, Zhongbo Tian, Hanjing Wang, Rui Wang, Ting Wang, Yi Wang, Baiting Wu, Jun Xu, Bowen Yang, Hui Wang, Weida Wang, Haochen Ye, Jiashuo Yu, Shan Yu, Xiaoyi Yu, Qirui Zeng, Qi Zhang, Ming Zhang, Wenwei Zhang, Bowen Zhou, Xinyu Zhou

B. Expert Activation Patterns

We further compare expert-activation pattern under two training recipes introduced in 3. As shown in 6, the Mobius-7B model trained from scratch (left) exhibits a comparatively uniform activation distribution across reasoning layers. Intern-S2-Mobius-35B (right), continually pre-trained from Qwen3.5-35B, activates experts over a wider range while retaining a pronounced block-diagonal pattern. The latter suggests that expert routing remains influenced by the layer-specific organization of the source checkpoint.

These observations suggest that continual pre-training can broaden expert utilization without fully removing the routing prior induced by architectural conversion. Training Mobius from scratch may therefore permit more flexible access to the shared expert pool; whether this translates into stronger model capability requires controlled evaluation at comparable scale.

C. Toy Task of Compositional Generalization

We select a Compositional Generalization task from Physics of LLM [1]. On this task, we compared Transformer against our yet-to-be-released, newer Mobius architecture. As shown in Fig 7, Mobius achieved substantially better convergence efficiency and final scores. This compositional generalization task uses single-hop knowledge of 500 entities and two-hop knowledge of 400 entities as the training set, and two-hop knowledge of the remaining 100 entities as the test set. By comparing test-set scores, we examine whether the model can genuinely learn the connections between different knowledge pieces and achieve compositional generalization.

D. Case Study: Chain-of-Thought Reasoning

We further present an example from the field of biology in 3. In this case, Mobius also delivers a mush shorter reasoning trace than Transformer.

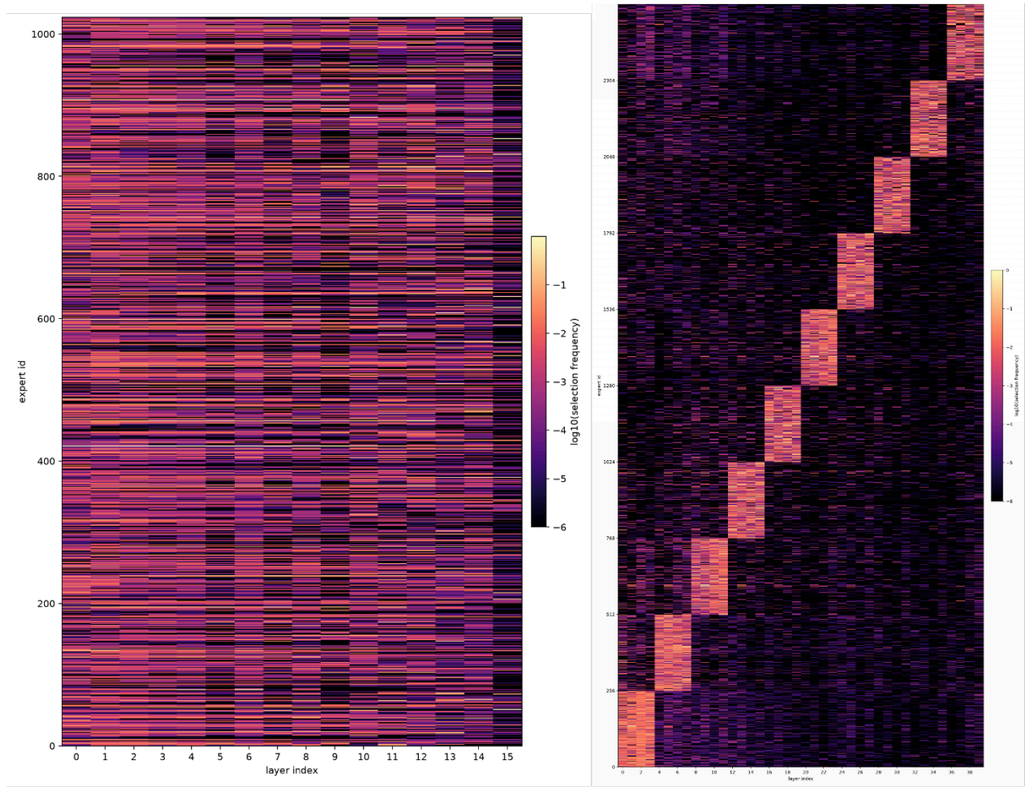


图6：两种训练方案下跨层的专家激活模式。左图展示了从头训练的Mobius-7B；右图展示了通过架构转换和持续预训练获得的Intern-S2-Mobius-35B。行表示专家ID，列表示层索引，颜色表示选择频率的以10为底的对数。右图中的专家ID经过重新排序，以揭示继承的路由结构。

E. 潜在推理的逐层分析

我们采用逐层预测视角来考察词元预测在潜在计算过程中如何演化。在相同的教师强制上下文下，我们对每一层的隐藏状态进行解码，以进行标准的下一个词元预测（$t+1$）以及后续四个MTP位置（$t+2$至$t+5$）的预测。如图8所示，Mobius在其中间层中展现出比基线更具可解释性、更贴合目标的预测。其逐层预测遵循更为连贯的语义轨迹，这表明任务相关信息被集中到紧凑内部表示中，而非分散在竞争词元候选之间。进一步的潜在迭代逐步优化该表示并改善后续位置的预测，最终生成完全被接受的五词元草稿。相比之下，基线模型的中间层预测稳定性较差，其草稿在第三个预测位置被拒绝，仅留下两个词元的接受前缀。该示例表明，Mobius可能形成更紧凑的内部表示，并在潜在空间中对其进行迭代优化，从而可能有利于多令牌预测。

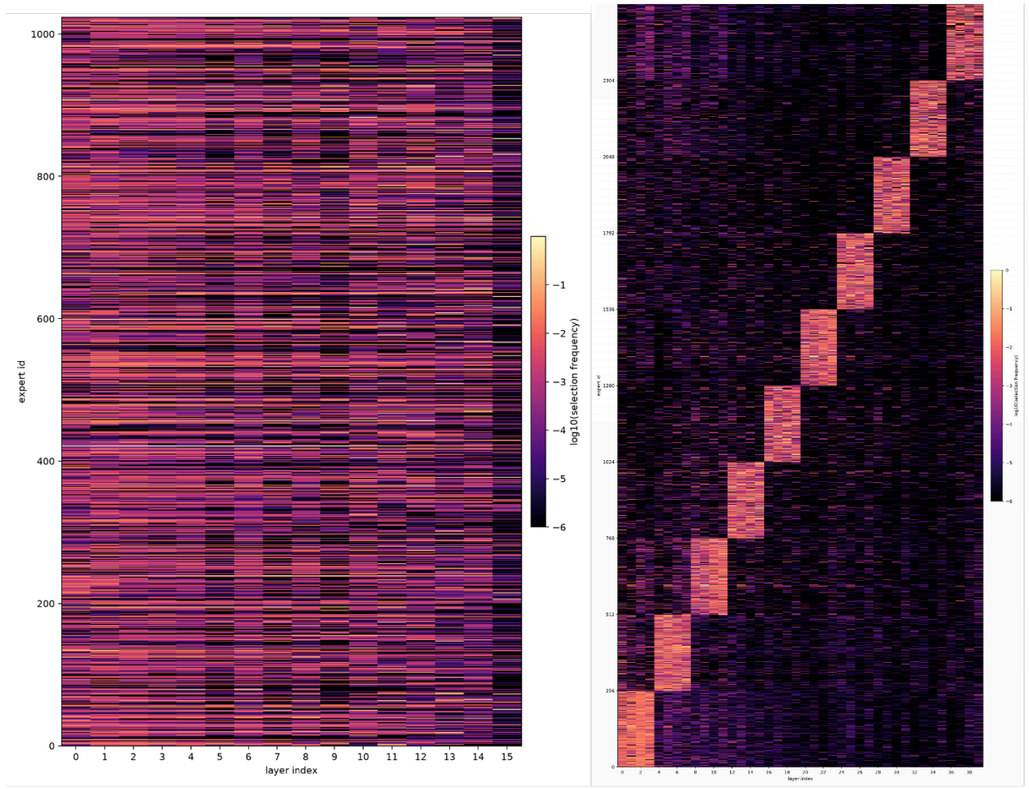


Figure 6: Expert-Activation pattern across layers under two training recipes. The left panel shows Mobius-7B trained from scratch; the right panel shows Intern-S2-Mobius-35B obtained by architecture conversion and continual pre-training. Rows denote expert IDs, columns denote layer indices, and color represents the base-10 logarithm of selection frequency. Expert IDs in the right panel are reordered to reveal the inherited routing structure.

E. Layerwise Analysis of Latent Reasoning

We use a layerwise prediction lens to examine how token predictions evolve during latent computation. Given the same teacher-forced context, we decode the hidden state at each layer for the standard next-token prediction ($t+1$) and four subsequent MTP positions ($t+2$ to $t+5$). As shown in Figure 8, Mobius exhibits more interpretable, target-aligned predictions in its intermediate layers than the baseline. Its layerwise predictions follow a more coherent semantic trajectory, suggesting that task-relevant information is concentrated into a compact internal representation rather than dispersed across competing token candidates. Further latent iterations progressively refine this representation and improve predictions at subsequent positions, ultimately yielding a fully accepted five-token draft. In contrast, the baseline exhibits less stable intermediate predictions, and its draft is rejected at the third predicted position, leaving only a two-token accepted prefix. This example suggests that Mobius may form more compact internal representations and refine them iteratively in latent space, potentially benefiting multi-token prediction.

（a）任务描述

（a）性能

图7：选自大语言模型物理学的组合泛化任务。Mobius的未来版本（未发布）比Transformer具有更快的收敛速度和更高的最终得分。训练集包含500个实体的单跳 + 400实体两跳，而测试集包含100个实体的两跳留出集。该任务评估模型是否学习跨知识连接，而不仅仅是记忆。

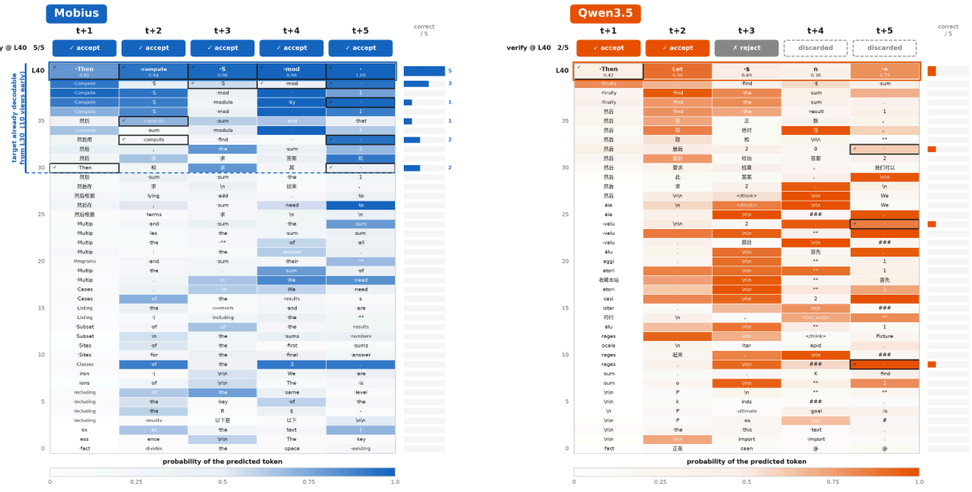
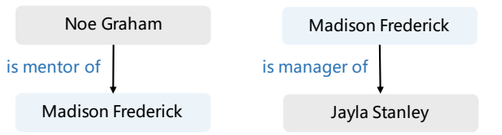


图8：在相同的教师强制上下文下，Mobius与Qwen3.5基线的逐层预测视角。$t+1$ 列表示标准的下一个词元预测，而 $t+2$ 至 $t+5$ 为后续的MTP预测。每个单元格显示从给定隐藏状态视图预测的词元；颜色强度表示其概率，黑色轮廓标记与目标延续的一致性。Mobius展现出更多目标对齐的中间预测，并生成完全被接受的五词元草稿，而基线仅产生两个词元的接受前缀。



(a) Task Description

(a) Performance

Figure 7: The Compositional Generalization task selected from Physics of LLM. A future-version of Mobius(unreleased) enjoys faster convergence, higher final scores than Transformer. The train set contains 500-entity single-hop + 400-entity two-hop, while the test set contains 100-entity two-hop holdout. This task evaluates whether models learn cross-knowledge connections, not mere memorization.

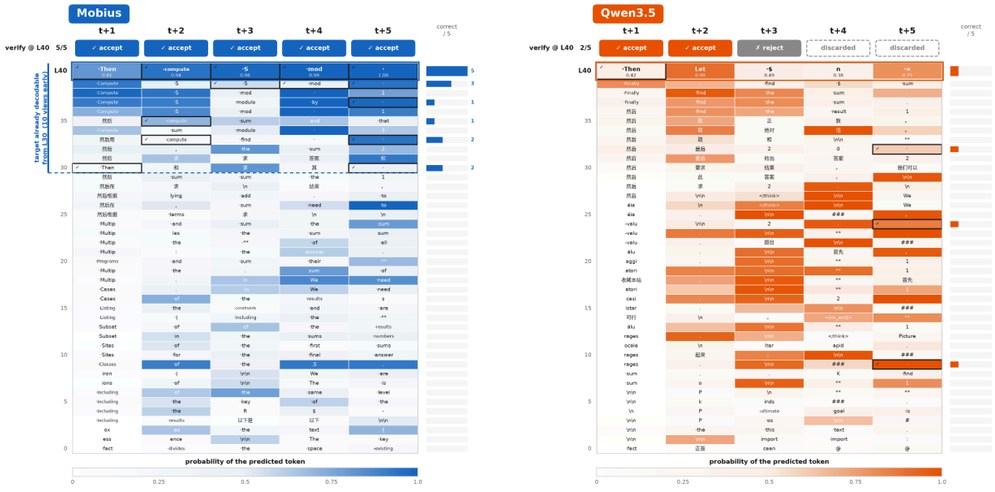


Figure 8: Layerwise prediction lens for Mobius and the Qwen3.5 baseline under an identical teacher-forced context. The $t+1$ column represents standard next-token prediction, while $t+2$ through $t+5$ are subsequent MTP predictions. Each cell shows the token predicted from a given hidden-state view; color intensity denotes its probability, and a black outline marks agreement with the target continuation. Mobius exhibits more target-aligned intermediate predictions and produces a five-token draft accepted in full, while the baseline produces only a two-token accepted prefix.

表3：Intern-Spin-35B与Qwen3.5-35B在光合作用多项选择题上的步骤对齐比较。两个模型均选择了正确答案（选项D）。令牌计数使用Qwen3.5-35B分词器计算。

推理轨迹对比

对齐步骤	Intern-Spin-35B	Qwen3.5-35B
任务框架	**36个词元**。 “用户正在询问关于 光合作用中光反应 与卡尔文循环之间的关系。 ”	**35个词元**。 “用户希望我回答……” 问题：光反应……为 卡尔文循环提供以下哪项 内容？”
核心生物学机制	**108个词元**。 “光依赖反应……产生 氧气、ATP 和 NADPH。” “卡尔文循环……利用 ATP 和 NADPH ……将 CO_2 固定为糖类。”	**276 个词元**。 “光依赖反应……输出： 氧气（O_2）、ATP、NADPH……卡尔文 循环……输入：CO_2、ATP、NADPH。 ” “……还原过程需要ATP和NADPH …… ”
选项评估	**240个词元**。 “D：ATP提供能量，NADPH 提供高能电子…… ” “H：ATP和NADPH在光反应中产生， 而非在那里被分解 … ” …	**543个词元**。 “D … 正确。光反应 产生ATP（能量）和NADPH （还原力/电子）。. .. ” “H .. . ATP和NADPH在 光反应中产生，并在 .. . 中消耗 卡尔文循环。”
重复推导	—	**205个词元**。 “结论：选项D准确 描述了……ATP与 NADPH……在卡尔文循环中被用于 固定碳。 ” **“逐步推导：**……光 反应产生ATP和NADPH → 卡尔文循环消耗它们。…… ”
重复选项检查	—	**143个词元**。 “最终检查：……选项A： CO_2 来源 是错误的……选项D：ATP/NADPH 转移是正确的……
推理闭合	**13个词元**。 “结论：选项D是正确 答案。”	**8个词元**。 “正确选项为D。 ”
可见的最终答案	**75个词元**。 “光反应……以ATP形式储存 和NADPH。这些分子随后 用于卡尔文循环中…… ” “答案：D”	**439个词元**。 “光合作用的过程分为 两个主要阶段……ATP和 NADPH……被运输到卡尔文 循环。” “… D：这表明ATP和NADPH是 … 被供应的。这是准确的……。” “答案：D”
总计	**472 个词元**	**1,649 个词元**

注：令牌计数是对每个原始预测的连续边际计数；因此，它们精确地加总为总输出长度。所显示的摘录是来自相应推理片段的代表性原始文本。省略号表示省略的原始文本。

Table 3: Step-aligned comparison between Intern-Spin-35B and Qwen3.5-35B on a photosynthesis multiple-choice question. Both models select the correct answer (Option D). Token counts are computed using the Qwen3.5-35B tokenizer.

Reasoning Trace Comparison

Aligned Step	Intern-Spin-35B	Qwen3.5-35B
Task framing	**36 tokens.** “The user is asking a question about the relationship between the light reactions and the Calvin cycle in photosynthesis.”	**35 tokens.** “The user wants me to answer … **Question:** The light reactions … supply the Calvin cycle with which of the following?”
Core biological mechanism	**108 tokens.** “Light-dependent reactions … producing oxygen, ATP, and NADPH.” “Calvin cycle … uses ATP and NADPH … to fix CO_2 into sugars.”	**276 tokens.** “Light-dependent reactions … Outputs: Oxygen (O_2), ATP, NADPH. … Calvin Cycle … Inputs: CO_2, ATP, NADPH.” “… reduction requires ATP and NADPH …”
Option evaluation	**240 tokens.** “D: ATP provides energy, NADPH provides high-energy electrons …” “H: ATP and NADPH are produced in light reactions, not broken down there …”	**543 tokens.** “D … Correct. The light reactions generate ATP (energy) and NADPH (reducing power/electrons). …” “H … ATP and NADPH are produced in the light reactions and consumed … in the Calvin cycle.”
Repeated derivation	—	**205 tokens.** “**Conclusion:** Option D accurately describes the relationship … ATP and NADPH … are used in the Calvin cycle to fix carbon.” “**Step-by-step derivation:** … Light Reactions produce ATP and NADPH → Calvin Cycle consumes them. …”
Repeated option check	—	**143 tokens.** “**Final check:** … Option A: CO_2 source is wrong. … Option D: ATP/NADPH transfer is correct. …”
Reasoning closure	**13 tokens.** “Conclusion: Option D is the correct answer.”	**8 tokens.** “The correct option is D.”
Visible final answer	**75 tokens.** “The light reactions … stored in ATP and NADPH. These molecules are then used in the Calvin cycle …” “ANSWER: D”	**439 tokens.** “The process of photosynthesis is divided into two main stages … ATP and NADPH … are transported to the Calvin cycle.” “… D: This states ATP and NADPH are supplied. This is accurate. …” “ANSWER: D”
Total	**472 tokens**	**1,649 tokens**

Note: The token counts are sequential marginal counts over each raw prediction; therefore, they sum exactly to the total output length. The displayed excerpts are representative original text from the corresponding reasoning segment. Ellipses indicate omitted original text.